L08 — study notes

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Lecture 8 — Signaling and perfect Bayesian equilibrium · Map of the notes

Estimated unit. Built on 2026-10-16 from the study guide before the lecture. Reconcile with the slides when Aula/Lecture08.pdf arrives (/notes 08 –reconcile).

One claim. Subgame perfection goes blind once types are private: the uninformed player always moves inside an information set, so the only proper subgame is the whole game. **Perfect Bayesian equilibrium restores sequential rationality** by adding beliefs — pinned by Bayes' rule on the path, free off it — and asking for a best reply at every information set. **In signaling games** the informed player's costly action can reveal his type, but only if it costs the types differently; the free off-path beliefs then support **continua** of pooling and separating equilibria. **The intuitive criterion** reads a deviation as coming from whoever could gain by it, and in every two-type game of this lecture leaves **only the least-cost separating equilibrium**.

Three layers per topic:

Layer	What it means here
U — intuition	Gibbons's prose and examples (Figs. 4.1.1, 4.2.2, Beer–Quiche), M&R's Spence story, the speeches of Tadelis p. 333
C — course (examinable)	Tadelis Ch. 15–16: Defs. 15.1–15.5, Requirements 15.1–15.4, Prop. 15.1, the MBA game, limit pricing, Def. 16.1; Gibbons §4.2's signaling requirements and Spence's separating/pooling sets
G — graduate	MWG §9.C–9.D (weak PBE, consistency, forward induction), §13.C's lemmas and welfare figure, 13.AA's dominance refinements. **Named, not examinable unless the lecture brings it**

Reading order

Part I — sequential rationality with incomplete information (Tadelis Ch. 15)

1. L08-01-subgame-perfection-fails — §15.1, pp. 303–307: the entry game with a weak and a competitive entrant; the Bayesian normal form at $p = \frac{1}{2}$ cell by cell; **two BNE, both subgame perfect**, because no proper subgame starts after entry; Gibbons's Fig. 4.1.1 as the same failure without Nature.
2. L08-02-beliefs-and-pbe — §15.2, pp. 307–312: Defs. 15.1–15.3, **Requirements 15.1–15.4**, Bayes' rule derived step by step; why **no belief** supports fighting; **Prop. 15.1 with its proof**; "weak" PBE vs Gibbons's Requirement 4.
3. L08-03-sequential-equilibrium — §15.3, pp. 312–313: Fig. 15.4's absurd PBE (U, R); **consistency** as the limit of Bayes under trembles; every SE is a PBE, not conversely.

Part II — signaling games (Tadelis Ch. 16, Gibbons §4.2)

4. L08-04-signaling-games — Ch. 16 opening, Gibbons §4.2.A: pooling / separating / hybrid, **Gibbons's Fig. 4.2.2 solved for all four candidates**, the five-step method; Myers–Majluf.
5. L08-05-mba-game — §16.1, pp. 319–323: the **16-cell matrix** (with one misprint found), the two BNE, the separating PBE (DU, BM) and the pooling PBE (UU, BB) **held up by** $\mu_D \leq \frac{3}{8}$; what changes with the prior.
6. L08-06-spence-job-market — Gibbons §4.2.B, M&R pp. 154–156: single crossing; **separating** $e_H \in [1.5, 3]$, pooling $e_p \in [0, 0.45]$, the **least-cost 1.5**; who gains ($p^* = 0.5$); hybrids connecting the two families.
7. L08-07-limit-pricing — §16.2, pp. 323–332: the low-cost incumbent **over-produces to be believed**: separating $q_1^{1L} \in [2.62, 3.32]$, pooling $\bar{q}_1 \in [1.493, 1.917]$, every bound derived (and one book misprint, $\frac{7}{36}$ not $\frac{7}{18}$).
8. L08-08-intuitive-criterion — §16.3, Gibbons §4.4: Def. 16.1 (with its v_2 typo corrected) as a four-step test; MBA pooling fails; limit-pricing pooling fails at $q = 3$; **Beer–Quiche**; in Spence **only** $e_H = 1.5$ **survives**.

Support

9. L08-09-formulas — every definition with its number, every number of every example, the study-guide items mapped.
10. L08-10-pitfalls — ten confusions with their counterexamples and **26 original True/False items** with justifications.
11. L08-11-python — Bayes' rule, an exhaustive pure-PBE solver for two-type signaling games, the intuitive-criterion test, Spence and limit pricing, **all asserted against the books**.

The one result to hold

PBE $= \left(\sigma^*, \mu\right)$: $\mu = \text{Bayes}(\sigma^*, p)$ on the path, μ free off it, σ^* sequentially rational given μ at every informa

> **Everything in the lecture is a consequence of where beliefs are free.** Free off-path beliefs kill non-credible threats that no belief can justify (entry game), sustain pooling equilibria by punishing deviations with pessimistic inferences (MBA, Spence, limit pricing), and are disciplined back by refinements: consistency (note 03) and the intuitive criterion (note 08).

And one condition for any signal to work — single crossing: the action must cost the type that wants to be recognized *less at the margin*. In the MBA game it reads $c_H \leq w_M - w_B \leq c_L$; in Spence $c_e(L, e) > c_e(H, e)$.

Numbers and cases to have memorized

Case	Number	Meaning
Entry game	$E[v_2(A)] = 1 > -\mu = E[v_2(F)]$	no belief justifies fighting: (OO, F) is BNE and SPE but not PBE

Case	Number	Meaning
MBA game	$\mu^* = \frac{3}{8}$; $2 \le 4 \le 5$	employer's promotion threshold; the separation constraints
MBA pooling	$\mu_D \in [0, \frac{3}{8}]$	the off-path belief that keeps everyone out of the MBA
Spence (M&R)	$e_H \in [1.5, 3]$; $e^* = 1.5$; $\bar{w} = 29$; $p^* = 0.5$	separating set; least-cost; pooled wage; above p^* H loses from signaling
Limit pricing	$[2.62, 3.32]$ vs monopoly 2; pooling $[1.493, 1.917]$	over-production to signal low cost; pooling deters nothing
Fig. 15.4	$\mu > \frac{2}{3}$ vs consistent $\mu = \frac{1}{2}$	the PBE sequential equilibrium removes

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Figures

Built by `figures/make_figures.py` (run: `python Notas/L08/figures/make_figures.py`; it ends with asserts on the plotted numbers).

Figure	Shows	Lives in
L08-entry-game	Fig. 15.2's tree with (EO, A) highlighted; the incumbent's payoff from A and F against the belief	L08-01-subgame-perfection-fails §2
L08-mba-tree	Fig. 16.1 redrawn with the separating PBE (DU, BM)	L08-05-mba-game §1
L08-mba-beliefs	employer's profit from M and B against μ; threshold $\frac{3}{8}$; each equilibrium's beliefs	L08-05-mba-game §3
L08-spence-separating	(e, w) plane with M&R's numbers: indifference lines, separating $[1.5, 3]$, pooling $[0, 0.45]$	L08-06-spence-job-market §3
L08-spence-welfare	H's and L's net pay with and without signaling against the prior; $p^* = 0.5$	L08-06-spence-job-market §5
L08-limit-pricing	every incentive constraint of §16.2 as a parabola against its deviation payoff	L08-07-limit-pricing §3
L08-intuitive-criterion	why pooling at $e_p = 0.3$ fails: the deviation band $(1.35, 2.4)$	L08-08-intuitive-criterion §5

Interactive companions

- The Education Signal (`companions/companion-spence-signaling.html`) — Spence with linear costs (M&R pp. 155–156 defaults): θ_L, θ_H, the prior, c_L, c_H move the separating interval, the

least-cost e^* (the page's answer), the pooling interval, the high type's gain from signaling and p^*; single crossing can be switched off. Linked from notes 06 and 08.

- The MBA Game (`companions/companion-mba-game.html`) — Tadelis §16.1 (p. 321 defaults): the prior, both study costs and both wages; the page **enumerates every pure PBE**, shows the supporting beliefs on the employer's profit-vs-belief chart, and applies the intuitive criterion. Linked from notes 05 and 08. (Beer–Quiche was the other candidate; it is Tadelis Exercise 16.8, so the book's own example was used instead.)
- Verification: `python Notas/L08/companions/check_companions.py` — Python vs page at the default, all-low and all-high vectors plus landmarks, a sweep of every control, and a syntax check.

Scope and ceiling

- **Examinable (C):** Tadelis Ch. 15–16 as listed above; Gibbons §4.2.A–B's requirements and the Spence pooling/separating construction; the MBA game and limit pricing with their numbers; the intuitive criterion applied to two types.
- **Named only (G):** MWG's weak PBE vs PBE, consistency in general, Prop. 9.C.2, forward induction (§9.D), MWG 13.AA's dominance hierarchy and the three-type Example 13.AA.1; hybrid equilibria beyond the worked example; Gibbons §4.2.D (monetary policy).
- **Out (CLAUDE.md scope):** cheap talk (Tadelis Ch. 18, Gibbons §4.3.A, Muñoz-Garcia Ch. 10), mechanism design and screening as a design problem (Tadelis Ch. 14, MWG 13.D, Muñoz-Garcia Ch. 9 Exercise 8), moral hazard, reputation games (Tadelis Ch. 17).
- **Estimated:** the sections and their weight follow the study guide; the lecture may skip §15.3 or the limit-pricing algebra. Reconcile when the slides arrive.

Course-wide position

Lecture	Relation to this one
Lecture 1 — dominance, IESDS	the deviation logic of the intuitive criterion is a dominance argument restricted to equilibrium payoffs
Lecture 2 — Nash, mixed strategies	best responses to beliefs; Cournot best replies reused in limit pricing
Lecture 3 — backward induction, SPE	**the concept that fails here** and that PBE extends to information sets
Lecture 4 — multistage games, bargaining	M&R Ch. 5 takes bargaining under private information (pp. 140–149) straight into adverse selection and signaling (pp. 149–157)
Lecture 5 — repeated games I	the monetary-policy signaling game (Gibbons §4.2.D) builds on its policy game
Lecture 6 — collusion, relational contracts	reputation as a signal over time (Tadelis Ch. 17, background)
Lecture 7 — Bayesian games, adverse selection, auctions	**BNE is the input** to PBE; the lemons problem is what a signal escapes
L08	**beliefs, PBE, signaling, refinements**

Rules packs: rules/07 — adverse selection and signaling · rules/game-theory.

Bibliography actually used in these notes

Book	Where
Tadelis (2013), Ch. 15–16 (pp. 303–338) — the spine	notes 01–05, 07–08
Gibbons (1992), §4.1–4.2 (pp. 175–210) and **§4.4** (pp. 233–244)	U layer of 01–02, 04, 06, 08
Milgrom & Roberts (1992), Ch. 5 "Signaling" (pp. 154–157)	the Spence numbers of 06, the companion's defaults, signaling vs screening
MWG (1995), §9.C–9.D (pp. 282–292), **§13.C** (pp. 450–459), **13.AA** (pp. 467–471)	G layer of 02, 03, 06, 08
Muñoz-Garcia & Toro-Gonzalez (2019), Ch. 9, 11	exercises only (by number and page)

Related project material

- `Listas/pratica-aula-01-02.md` item 18 — M&R Food for Thought 5.2 (p. 164), credibility of signals of intent; solution `Listas/pratica-aula-01-02-solutions/item18.tex`.
- `Map/exercises-index.md` — MWG 13.C.1, 13.C.2, 13.C.4, 13.C.5, 13.C.6 (pp. 474–476), 13.AA.1 (p. 476); M&R Q 5.2 (p. 164).
- `Map/books-index.md` — Tadelis Ch. 15–16 and Gibbons §4.2 section lists.
- **After-class practice (study guide):** Muñoz-Garcia Ch. 9 (pp. 375–451), Ch. 11 (pp. 481–520); MWG 13.C.1, 13.C.5, then 13.C.2, 13.C.4.
- No list, quiz or video for Lecture 8 exists yet.

15.1 · Why subgame perfection has no bite once types are private

Claim. Subgame perfection removes non-credible threats by asking for a Nash equilibrium in every *proper subgame*. Add one move by Nature that the second mover cannot see, and the second mover's decision nodes are tied together in one information set: **the only proper subgame is the whole game**, so subgame perfection collapses into Bayesian Nash equilibrium and the incredible threat survives. Sequential rationality has to be imposed **at information sets**, which needs beliefs — the next note.

Prev: — · Next: L08-02-beliefs-and-pbe · Map: L08-00-map

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1 · U — the benchmark: entry with complete information (Fig. 15.1, pp. 303–304)

An entrant (player 1) chooses Out (O) or Enter (E); after entry the incumbent (player 2) chooses Fight (F) or Accommodate (A). Payoffs (entrant, incumbent): $O \to (0, 2)$; $(E, F) \to (-1, -1)$; $(E, A) \to (1, 1)$.

	F	A
O	$0,\ 2$	$0,\ 2$
E	$-1,\ -1$	$1,\ 1$

Nash equilibria. (O, F): given F, O (0) beats E (-1); given O, the incumbent is indifferent ($2 = 2$), so F is a best response. (E, A): given A, E (1) beats O (0); given E, A (1) beats F (-1). Two pure equilibria.

Backward induction (Lecture 3): after E the incumbent compares $1 > -1$ and accommodates; anticipating this the entrant compares $1 > 0$ and enters. **Unique SPE** (E, A); (O, F) rests on a threat the incumbent would never carry out.

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2 · C — add a type: the entry game with incomplete information (Fig. 15.2, pp. 304–305)

Timing (p. 304):

1. Nature draws the entrant's type $\theta_1 \in \{W, C\}$ (weak, competitive), $\Pr\{\theta_1 = C\} = p$. The entrant learns it; the incumbent knows only p.
2. The entrant chooses E or O; the incumbent observes the choice.
3. After entry the incumbent chooses A or F **without knowing the type**.

Payoffs (entrant, incumbent):

Type	O	E then F	E then A
C (prob. p)	$0,\ 2$	$-1,\ -1$	$1,\ 1$
W (prob. $1-p$)	$0,\ 2$	$-2,\ 0$	$-1,\ 1$

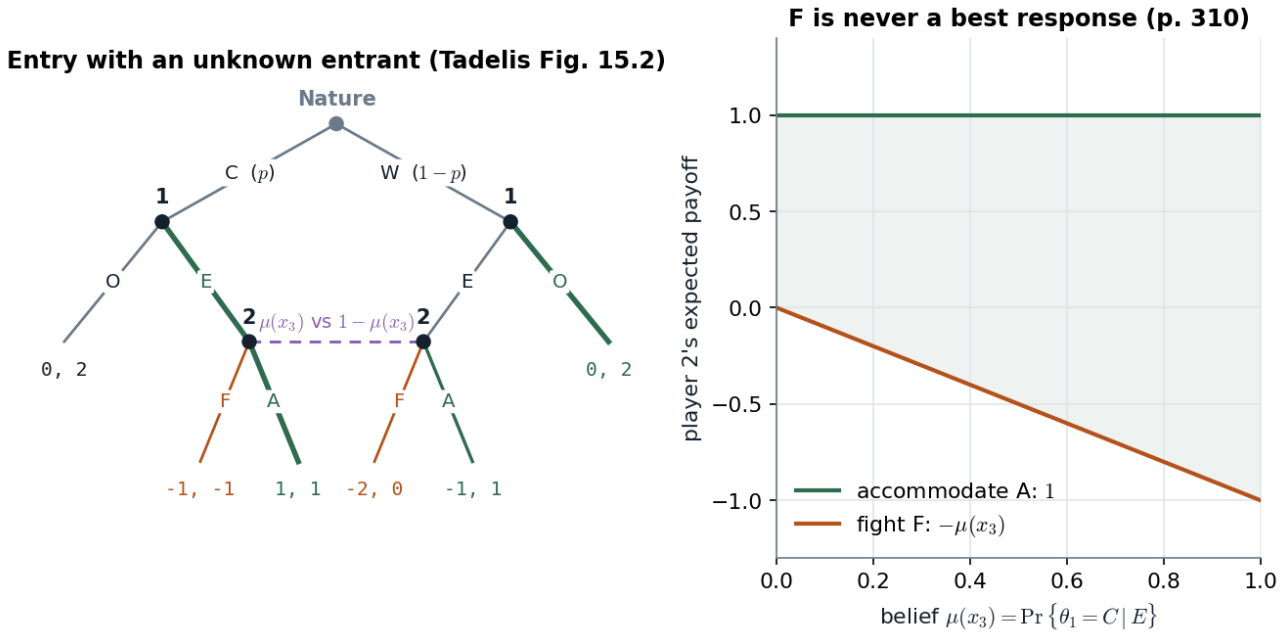


Figure — the entry game of Fig. 15.2 with the separating equilibrium (EO, A) highlighted (left), and the incumbent's expected payoff after entry as a function of its belief (right): A beats F by at least 1 at every belief. Source: Tadelis Fig. 15.2, p. 305, and the argument on p. 310.

The dashed line is the whole difficulty: after E the incumbent does not know which of x_3 (a competitive entrant) or x_4 (a weak one) it is at. On the right, however, the belief does not matter: A pays 1 against either type while F pays -1 or 0, so *every* belief makes A the best reply — which is exactly why the threat to fight is not credible (note 02, §5).

2.1 · Strategies are type-contingent

A strategy for the entrant is a plan for **each** type: $s_1 = s_1^C s_1^W$ with $s_1^\theta \in \{O, E\}$, so

$$S_1 = \{OO,\ OE,\ EO,\ EE\}, \qquad S_2 = \{F, A\}$$

(OE = "out if competitive, enter if weak"). The incumbent has one information set (after E) with two actions.

2.2 · The Bayesian normal form, cell by cell (p. 305)

Each cell averages over Nature. (OE, A)**:** with prob. p the type is C, plays O, payoffs $(0, 2)$; with prob. $1-p$ the type is W, plays E, the incumbent plays A, payoffs $(-1, 1)$:

$$Ev_1 = p \cdot 0 + (1-p)(-1) = p - 1, \qquad Ev_2 = p \cdot 2 + (1-p) \cdot 1 = 1 + p.$$

(The book prints $Ev_2 = 1 - p$ on p. 305; at $p = 1/2$ its own matrix shows $3/2 = 1 + p$, which is the correct value.)

(EE, F)**:** both types enter and are fought: $Ev_1 = p(-1) + (1-p)(-2) = p - 2$, $Ev_2 = p(-1) + (1-p) \cdot 0 = -p$.

At $p = \frac{1}{2}$ (p. 306):

	F	A
OO	$0,\ 2$	$0,\ 2$
OE	$-1,\ 1$	$-\frac{1}{2},\ \frac{3}{2}$
EO	$-\frac{1}{2},\ \frac{1}{2}$	$\frac{1}{2},\ \frac{3}{2}$
EE	$-\frac{3}{2},\ -\frac{1}{2}$	$0,\ 1$

Check two cells. (EO, F): C enters and is fought, W stays out: $v_1 = \frac{1}{2}(-1) + \frac{1}{2}(0) = -\frac{1}{2}$, $v_2 = \frac{1}{2}(-1) + \frac{1}{2}(2) = \frac{1}{2}$. (EE, A): $v_1 = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0$, $v_2 = \frac{1}{2}(1) + \frac{1}{2}(1) = 1$.

2.3 · The pure Bayesian Nash equilibria

Best responses of player 1: against F the column is $(0, -1, -\frac{1}{2}, -\frac{3}{2})$, maximum 0 at OO; against A it is $(0, -\frac{1}{2}, \frac{1}{2}, 0)$, maximum at EO. Best responses of player 2: row OO — tie $(2 = 2)$, both; OE — A $(\frac{3}{2} > 1)$; EO — A $(\frac{3}{2} > \frac{1}{2})$; EE — A $(1 > -\frac{1}{2})$. Mutual best responses:

$$\boxed{(OO, F) \quad \text{and} \quad (EO, A)} \qquad \text{(p. 306)}$$

> **The two BNE mirror the two NE of the complete-information game**: (OO, F) is the threat that keeps everyone out; (EO, A) is accommodation, under which the competitive type enters $(1 > 0)$ and the weak type stays out $(0 > -1)$.

Footnote 1 (p. 306): OE is strictly dominated by OO and EE by EO (the weak type always loses by entering: -2 or -1 against 0). There is also a continuum of mixed BNE in which the incumbent fights with probability $\varphi \geq \frac{1}{2}$ and the entrant plays OO: the competitive type then gets $\varphi(-1) + (1 - \varphi)(1) = 1 - 2\varphi \leq 0$ from entering, so staying out is a best response exactly when $\varphi \geq \frac{1}{2}$.

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3 · C — why subgame perfection cannot remove (OO, F) (p. 306)

(OO, F) has the same flaw as (O, F): **if the incumbent finds itself after entry, A is a strict best response** (§2's figure, right panel). Yet:

> **Subgame-perfect equilibrium requires a Nash equilibrium in every *proper* subgame. A proper subgame must start at a singleton information set and cut no information set. The incumbent's nodes x_3, x_4 form one information set, and the entrant's nodes hang from Nature's move, so the only proper subgame is the whole game** — and both BNE survive as SPE.

Concept	Removes (OO, F)?	Why
Nash / Bayesian Nash	no	F is never *played* on the path, so it costs the incumbent nothing
Subgame-perfect	**no**	no proper subgame begins after entry
Perfect Bayesian (note 02)	**yes**	it asks for a best reply **at the information set**, given a belief

The structural point (p. 307): with incomplete information, **every** move of the uninformed player is

made inside an information set that pools the informed player's types, so this failure "will carry over to all games of incomplete information."

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4 · U — the same failure without Nature (Gibbons Fig. 4.1.1, pp. 175–177)

The problem is not Nature as such; it is **non-singleton information sets**. Gibbons's game: player 1 chooses L, M or R; R ends the game at $(1, 3)$; after L or M player 2 knows only that R was not chosen and picks L' or R'. Normal form (Fig. 4.1.2):

	L'	R'
L	2, 1	0, 0
M	0, 2	0, 1
R	1, 3	1, 3

Pure NE: (L, L') and (R, R'). The game has **no proper subgames**, so both are subgame perfect. But give player 2 a belief $p = \Pr(\text{node after } L)$:

$$E[u_2(R')] = p \cdot 0 + (1-p) \cdot 1 = 1 - p, \qquad E[u_2(L')] = p \cdot 1 + (1-p) \cdot 2 = 2 - p.$$

$2 - p > 1 - p$ for **every** p: R' is never a best reply at the information set, so (R, R') is killed by Gibbons's Requirements 1–2 alone (p. 177). The same move — *a belief at every information set, and optimality given it* — is what note 02 formalizes.

Links

- Next: beliefs, Bayes' rule and the definition of PBE → L08-02-beliefs-and-pbe.
- Backward induction and subgame perfection: Lecture 3.
- Bayesian games, type-contingent strategies and BNE: Lecture 7.
- Rules: rules/game-theory items 2–3 (name the concept; check multiplicity).
- Exercises (number and page only): Tadelis 15.1 (p. 314); Gibbons Problem 4.1 (p. 245); Muñoz-Garcia Ch. 9 Exercise 1 (p. 376).

15.2 · Beliefs at every information set, Bayes' rule where it applies, best replies everywhere

Claim. A perfect Bayesian equilibrium is a strategy profile **plus a belief system**. Beliefs are **pinned by Bayes' rule wherever the equilibrium reaches an information set** and are **free where it does not**; given the beliefs, every player best-responds **at every information set**, reached or not. The freedom off the path is the concept's power (it kills threats no belief can justify) and its weakness (it can support almost anything — notes 03 and 08).

Prev: L08-01-subgame-perfection-fails · Next: L08-03-sequential-equilibrium · Map: L08-00-map

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1 · C — on and off the equilibrium path (Definition 15.1, p. 307)

Definition 15.1. Let σ^* be a Bayesian Nash equilibrium profile. An information set is **on the equilibrium path** if, given σ^* and the type distribution, it is reached with positive probability, and **off the equilibrium path** if it is reached with probability zero.

The definition needs a **candidate profile** first. In the entry game of L08-01-subgame-perfection-fails:

Entrant plays	Incumbent's information set after E	Reached with prob.
EO	on the path	$p \cdot 1 + (1-p) \cdot 0 = p > 0$
OO	**off** the path	$p \cdot 0 + (1-p) \cdot 0 = 0$

The informed player's strategy decides which of the uninformed player's information sets are reached (p. 308).

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2 · C — a system of beliefs (Definition 15.2, Requirement 15.1, p. 308)

Definition 15.2. A system of beliefs μ assigns to every information set h a probability distribution over its decision nodes: $\mu(x) \in [0,1]$ for $x \in h$, and $\sum_{x \in h} \mu(x) = 1$ for every $h \in H$.

Requirement 15.1. Every player has a well-defined belief over where he is in each of his information sets.

In the entry game the entrant's information sets are singletons (trivial beliefs); the incumbent holds $\mu(x_3) = \Pr\{C \mid E\}$ and $1 - \mu(x_3) = \Pr\{W \mid E\}$.

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3 · C — Bayes' rule on the path (Requirement 15.2, p. 309)

Let type C enter with probability σ_C and type W with σ_W (pure strategies are the cases $\sigma \in \{0,1\}$). Write the event "competitive and entered" and apply the definition of conditional probability:

$$\mu(x_3) = \Pr\{C \mid E\} = \frac{\Pr\{C \text{ and } E\}}{\Pr\{E\}} = \frac{\Pr\{C\}\Pr\{E \mid C\}}{\Pr\{C\}\Pr\{E \mid C\} + \Pr\{W\}\Pr\{E \mid W\}} = \frac{p\,\sigma_C}{p\,\sigma_C + (1-p)\,\sigma_W}.$$

Entrant's strategy	σ_C	σ_W	$\mu(x_3)$
EO	1	0	$\frac{p}{p} = 1$
OE	0	1	0
EE	1	1	p — **pooling teaches nothing**
OO	0	0	$\frac{0}{0}$: **undefined**
$\sigma_C = \frac{1}{2},\ \sigma_W = \frac{1}{4},\ p = \frac{1}{3}$			$\frac{1/6}{1/6 + 1/6} = \frac{1}{2}$

> **Requirement 15.2.** Given a BNE profile σ^*, beliefs at every information set **on the equilibrium path** are consistent with Bayes' rule.

Two constraints enter the formula (pp. 308–309): an **exogenous** one (Nature's prior p) and an **endogenous** one (the conjectured strategy σ). Beliefs are "correct" in the same sense in which Nash equilibrium makes beliefs about strategies correct.

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4 · C — off the path anything goes; then best-respond (Requirements 15.3–15.4, p. 310)

> **Requirement 15.3.** At information sets **off** the equilibrium path, any belief can be assigned, since Bayes' rule does not apply.

> **Requirement 15.4.** Given their beliefs, players' strategies are sequentially rational: at every information set h of player i,
>
> $$E[v_i(\sigma_i^*, \sigma_{-i}^*, \theta_i) \mid h, \mu] \;\geq\; E[v_i(s_i, \sigma_{-i}^*, \theta_i) \mid h, \mu] \quad \text{for all } s_i \in S_i.$$

> **Definition 15.3.** A BNE profile σ^* together with a system of beliefs μ is a **perfect Bayesian equilibrium** if they satisfy Requirements 15.1–15.4.

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5 · Worked — the entry game, settled (pp. 310–311)

Candidate (OO, F). The information set after E is off the path, so by Requirement 15.3 any $\mu(x_3) \in [0,1]$ is allowed. Requirement 15.4 asks whether F is a best reply to *some* belief:

$$E[v_2(A) \mid \mu] = \mu \cdot 1 + (1-\mu)\cdot 1 = 1, \qquad E[v_2(F) \mid \mu] = \mu(-1) + (1-\mu)\cdot 0 = -\mu.$$

$1 > -\mu$ for every $\mu \in [0,1]$ (the gap is at least 1: right panel of the figure in L08-01-subgame-perfection-fails §2). **No belief supports** F, so (OO, F) is **not** a PBE.

Candidate (EO, A). Every information set is reached; Bayes gives $\mu(x_3) = 1$ (only C enters). Check each information set:

Who, where	Belief	Prescribed	Payoff	Alternative	Payoff
Incumbent after E	$\mu = 1$	A	1	F	-1
Entrant, type C	—	E	1 (then A)	O	0
Entrant, type W	—	O	0	E	-1 (then A)

All strict: $\boxed{(EO, A),\ \mu(x_3) = 1}$ **is the unique pure PBE.**

> **What did the work was not Bayes' rule — the belief was unrestricted — but the requirement to *have* a belief and to best-respond to it at an unreached information set.**

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6 · C — Proposition 15.1: when every information set is reached (p. 311)

> **Proposition 15.1.** If a (possibly mixed) profile σ^* is a BNE of a Bayesian game Γ and induces **every** information set to be reached with positive probability, then σ^* together with the belief system derived from σ^* and the type distribution by Bayes' rule is a PBE of Γ.

Proof, every step. Suppose not: σ^* is a BNE, all information sets are reached, but (σ^*, μ) is not a PBE.

1. Requirements 15.1–15.3 hold by construction: every information set has a belief, and each is given by Bayes' rule because each is on the path ($\Pr\{h\} > 0$, so the denominator is positive).
2. So Requirement 15.4 must fail: some player i at some information set h is not best responding to $\mu(h)$.
3. Because h is reached with positive probability, player i could switch to the better action at h and **raise his ex-ante expected payoff** by $\Pr\{h\} \times (\text{gain at } h) > 0$.
4. That is a profitable deviation from σ^* in the Bayesian game — contradicting that σ^* is a BNE. ■

> **Off-path information sets are the *only* place where BNE and PBE can differ.** A BNE's only licence to play badly is at information sets it never reaches, because there a bad action costs nothing ex ante. PBE withdraws that licence.

How to use it (p. 311). (i) Find the BNE (the matrix of L08-01-subgame-perfection-fails §2.3); (ii) for each, compute beliefs on the path by Bayes; (iii) if nothing is off the path, Prop. 15.1 finishes; otherwise (iv) look for an off-path belief that makes the prescribed off-path action a best reply. In signaling games one usually works directly: **guess a sender strategy, derive beliefs, derive the receiver's best replies, pick supporting off-path beliefs, check the sender's incentive constraints** (note 04, §3).

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7 · G — "weak" PBE, and Bayes' rule *where possible* off the path (Remark, Fig. 15.3, pp. 311–312)

Requirement 15.3 puts **no** restriction off the path, which is why some texts call Tadelis's concept **weak perfect Bayesian equilibrium** (MWG Def. 9.C.3, p. 285). A stronger version asks for Bayes' rule **"where possible"** even off the path.

Fig. 15.3: player 1 chooses L (to player 2) or R (to player 3); players 2 and 3 each choose between two actions (A/B and C/D); player 4 then moves at two information sets, one after player 2's move and

one after player 3's, not seeing which action was taken. Let the profile have 1 play L, 2 play A with probability σ_A and 3 play C with probability σ_C.

- Player 4's information set after player 2 is on the path: $\mu_A = \sigma_A$ by Bayes.
- The one after player 3 is off the path (1 never plays R): Requirement 15.3 allows anything. But if player 4 finds himself there, **only player 1 has deviated**; there is no reason to think player 3 deviated too, so the natural belief is $\mu_C = \sigma_C$.

Gibbons builds this into his definition (Requirement 4, p. 180: *"At information sets off the equilibrium path, beliefs are determined by Bayes' rule and the players' equilibrium strategies where possible"*), and shows with Fig. 4.1.4 (pp. 180–181) why it matters: there the strategies (A, L, L') with belief $p = 0$ satisfy his Requirements 1–3 and are a Nash equilibrium, **but not subgame perfect** — player 3's belief contradicts player 2's strategy L, and Requirement 4 forces $p = 1$, after which player 3 must play R'.

Gibbons (pp. 177–180)	Tadelis (pp. 308–310)	Content
Requirement 1	Requirement 15.1 (+ Def. 15.2)	a belief at every information set
Requirement 2	Requirement 15.4	sequential rationality given beliefs
Requirement 3	Requirement 15.2	Bayes' rule on the path
Requirement 4	— (Tadelis's 15.3 is weaker)	Bayes' rule off the path **where possible**

> **In a two-stage signaling game the distinction is moot**: the receiver's off-path information sets follow messages no type sends, so Bayes' rule is never "possible" there — Gibbons notes Requirement 4 is vacuous in signaling games (p. 186). Everything examinable in notes 04–08 is the same under both definitions.

MWG (G). Def. 9.C.1 (p. 283) is the belief system; Def. 9.C.2 (p. 284) sequential rationality via $E[u_i \mid H, \mu, \sigma_i, \sigma_{-i}]$; Def. 9.C.3 (p. 285) the weak PBE. Not examinable beyond what Tadelis states.

Links

- Prev: L08-01-subgame-perfection-fails · Next: L08-03-sequential-equilibrium.
- The signaling-game version of these requirements: L08-04-signaling-games §2.
- Bayesian games and BNE (the input to Prop. 15.1): Lecture 7.
- Rules: rules/game-theory item 9 ("beliefs consistent with Bayes' rule wherever possible").
- Code: `posterior` and the entry-game check in L08-11-python §1–2.
- Exercises (number and page only): Tadelis 15.1–15.2 (p. 314), 15.3 (p. 315), 15.4–15.5 (p. 316); Gibbons Problems 4.1–4.2 (p. 245); Muñoz-Garcia Ch. 9 Exercise 1 (p. 376).

15.3 · Sequential equilibrium: discipline the beliefs nobody should be able to choose

> **Claim.** Requirement 15.3 lets off-path beliefs be anything, and some games then have PBE that rest on beliefs **no sequence of slightly-mistaken plays could ever generate**. Kreps and Wilson's **consistency** asks that beliefs be the **limit of Bayes' rule applied to totally mixed strategies converging to the equilibrium**. Sequential equilibrium = consistent PBE. It is the first *refinement*; the intuitive criterion (note 08) is the one signaling games need.

Prev: L08-02-beliefs-and-pbe · Next: L08-04-signaling-games · Map: L08-00-map

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1 · C — a game with an unreasonable PBE (Fig. 15.4, pp. 312–313)

Nature picks one of two nodes with probability $\frac{1}{2}$ each; **player 1 does not observe the draw** and chooses U (ending the game at $(1,0)$) or D. After D player 2 moves at an information set $\{x_1, x_2\}$ (one node per draw) and chooses L or R. Payoffs after D (player 1, player 2):

Node	L	R
x_1	2, 2	0, 3
x_2	2, 2	0, 0

Player 2 with belief $\mu = \mu_2(x_1)$:

$$E[v_2(L)] = 2\mu + 2(1-\mu) = 2, \qquad E[v_2(R)] = 3\mu + 0(1-\mu) = 3\mu.$$

R is a best reply iff $3\mu \geq 2$, i.e. $\mu \geq \frac{2}{3}$.

PBE 1 — (D, L) with $\mu = \frac{1}{2}$. D is played, so the information set is on the path and Bayes' rule gives

$$\mu_2(x_1) = \frac{\frac{1}{2}\Pr\{D\}}{\frac{1}{2}\Pr\{D\} + \frac{1}{2}\Pr\{D\}} = \frac{1}{2}$$

(player 1 cannot condition on Nature, so $\Pr\{D\}$ is the same at both nodes). Then $3 \cdot \frac{1}{2} = \frac{3}{2} < 2$: L is the best reply; given L, player 1 compares $D \to 2$ with $U \to 1$ and plays D. ✓

PBE 2 — (U, R) with $\mu > \frac{2}{3}$. U is played, the information set is off the path, Requirement 15.3 allows any belief; pick $\mu_2(x_1) > \frac{2}{3}$, so $3\mu > 2$ and R is a best reply; given R, player 1 compares $U \to 1$ with $D \to 0$ and plays U. ✓

> **PBE 2 is absurd:** whatever made player 1 play D by mistake, he could not have made it depend on a draw he never saw, so the two nodes must stay equally likely. Yet nothing in Requirements 15.1–15.4 says so.

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2 · C — consistency and sequential equilibrium (Def. 15.4–15.5, p. 313)

Definition 15.4. A profile σ^* together with a belief system μ^* is **consistent** if there is a sequence of **non-degenerate** (totally mixed) strategy profiles $\{\sigma^k\}_{k=1}^{\infty}$ and the sequence of beliefs $\{\mu^k\}$ derived from each σ^k by Bayes' rule, such that $\lim_{k\to\infty}(\sigma^k, \mu^k) = (\sigma^*, \mu^*)$.

Definition 15.5. A profile σ^* with beliefs μ^* is a **sequential equilibrium** if it is a consistent perfect Bayesian equilibrium.

Applying it to Fig. 15.4. In any totally mixed σ^k, player 1 plays D with some $\varepsilon_k \in (0,1)$ — the same at both nodes. So for every k

$$\mu_2^k(x_1) = \frac{\frac{1}{2}\varepsilon_k}{\frac{1}{2}\varepsilon_k + \frac{1}{2}\varepsilon_k} = \tfrac{1}{2}, \qquad \text{hence} \qquad \lim_k \mu_2^k(x_1) = \tfrac{1}{2}.$$

ε_k	$\mu_2^k(x_1)$
0.1	$0.05/0.10 = \frac{1}{2}$
0.001	$0.0005/0.001 = \frac{1}{2}$
10^{-6}	$\frac{1}{2}$

The only consistent belief is $\frac{1}{2} < \frac{2}{3}$, under which R is not a best reply. (U, R) **is not a sequential equilibrium;** (D, L) **with** $\mu = \frac{1}{2}$ **is the unique one.**

> **Consistency turns "off-path beliefs are free" into "off-path beliefs are what trembles would teach you."** The trembles must be *the players' own* — here, one tremble by a player who cannot see Nature's draw, which therefore cannot shift the odds between the two nodes.

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3 · C — where the concepts sit

every sequential equilibrium is a PBE (Def. 15.5; the converse fails: Fig. 15.4's (U, R))

Concept	Beliefs required	Off-path beliefs	Kills in Fig. 15.4
Bayesian Nash	none	—	nothing
PBE (Tadelis, "weak")	Bayes on the path	anything	nothing ((U, R) survives)
Sequential	limits of Bayes from trembles	**pinned** to $\frac{1}{2}$	(U, R)

Tadelis's practical verdict (p. 313): sequential equilibrium "is a bit harder to apply", so applications use PBE; and **even sequential equilibrium can admit unreasonable outcomes**, which motivates the signaling-specific refinement of L08-08-intuitive-criterion.

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4 · G — the formal side (MWG §9.C)

- **MWG Def. 9.C.4 (p. 290):** sequential equilibrium = sequentially rational (Def. 9.C.2) and consistent in the Kreps–Wilson sense — the same as Tadelis Def. 15.4–15.5.
- **MWG Prop. 9.C.2 (p. 291):** *"In every sequential equilibrium (σ, μ) of an extensive form game Γ_E, the equilibrium strategy profile σ constitutes a subgame perfect Nash equilibrium of Γ_E."* (Weak PBE does **not** guarantee this — Gibbons Fig. 4.1.4, note 02 §7.)
- **Existence (Gibbons footnote 1, p. 179):** Kreps and Wilson show every **finite** game has a sequential equilibrium, hence a perfect Bayesian equilibrium.
- **MWG §9.D "Reasonable Beliefs and Forward Induction" (p. 292)** opens the programme of further restrictions on off-path beliefs that Chapter 13's Appendix A applies to signaling — the bridge to L08-08-intuitive-criterion §5. Named only.

Links

- Prev: L08-02-beliefs-and-pbe · Next: L08-04-signaling-games.
- The refinement that matters for signaling: L08-08-intuitive-criterion.
- Subgame perfection, which sequential equilibrium implies: Lecture 3.
- Code: the tremble computation in L08-11-python §8.
- Exercises (number and page only): Tadelis 15.6 "Selten's Horse" (p. 317).

16 · 4.2.A · Signaling games: who moves, what is learned, and the four-step method

Claim. A signaling game is the simplest dynamic game of incomplete information: an informed **Sender** moves first, an uninformed **Receiver** sees the move and responds. Its PBE come in three kinds — **pooling** (the message says nothing; posterior = prior), **separating** (the message reveals the type), **hybrid** (it reveals some). Every candidate is checked the same way: **guess the Sender's strategy → Bayes on the path → Receiver's best replies → off-path beliefs that keep the Sender in line → Sender's incentive constraints.**

Prev: L08-03-sequential-equilibrium · Next: L08-05-mba-game · Map: L08-00-map

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1 · U — why signals have to be costly (pp. 318–319)

A strong entrant would like to tell the incumbent "I am strong, don't waste money fighting me". But a weak entrant would like to say the same thing, so **saying it** proves nothing. A signal works only if it is **an action that costs the types differently** — Spence's (1973) insight: education can raise wages *without* raising productivity, purely because it is cheaper for the able to acquire. (Costless messages — cheap talk, Tadelis Ch. 18 — are out of this course's scope.)

Tadelis's four components (p. 318):

1. **Nature** draws player 1's type, which player 2 does not know but **cares about** (common values).
2. Player 1 has a **rich action set**: at least as many actions as types, and **each action costs each type differently**.
3. Player 1 moves first; player 2 responds **after observing** player 1's choice.
4. Player 2 **updates** his belief from player 1's choice (given his belief about player 1's strategy) and best-responds to the updated belief.

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2 · C — the formal game and the PBE requirements (Gibbons pp. 183–188)

Types $T = \{t_1, \dots, t_I\}$ with prior $p(t_i) > 0$; messages $M = \{m_1, \dots, m_J\}$; actions $A = \{a_1, \dots, a_K\}$. A pure strategy for the Sender is a **function** $m(t_i)$; for the Receiver, a **function** $a(m_j)$. With two types, two messages and two actions each side has four pure strategies (p. 186):

Sender strategy	t_1 sends	t_2 sends	Kind
1	m_1	m_1	**pooling**
2	m_1	m_2	**separating**
3	m_2	m_1	**separating**
4	m_2	m_2	**pooling**

The Sender's information sets are singletons, so Requirement 1 bites only on the Receiver:

Signaling Requirement 1. After any message m_j the Receiver has a belief $\mu(t_i \mid m_j) \geq 0$ with $\sum_{t_i \in T} \mu(t_i \mid m_j) = 1$.

2R. For each m_j, $a^*(m_j)$ maximizes $\sum_{t_i} \mu(t_i \mid m_j)\, U_R(t_i, m_j, a_k)$.

2S. For each t_i, $m^*(t_i)$ maximizes $U_S(t_i, m_j, a^*(m_j))$.

3. If some type sends m_j in equilibrium (let $T_j = \{t_i : m^*(t_i) = m_j\} \neq \emptyset$), the belief after m_j follows Bayes' rule: $\mu(t_i \mid m_j) = \dfrac{p(t_i)}{\sum_{t_i \in T_j} p(t_i)}$ for $t_i \in T_j$.

Definition (p. 188). A pure-strategy PBE of a signaling game is $m^*(t_i)$, $a^*(m_j)$ and $\mu(t_i \mid m_j)$ satisfying Requirements (1), (2R), (2S) and (3).

These are Tadelis's Requirements 15.1–15.4 specialized; Requirement 4 ("Bayes where possible off the path") is vacuous here, because an unused message is reached by no strategy (p. 186).

What each kind implies for beliefs (Tadelis pp. 318–319):

Kind	On-path beliefs	Off-path beliefs	What holds it together
Pooling	= the prior	at every unused message: free, but must support the Receiver's reply	the Receiver's reply to the unused message deters deviations
Separating	0 or 1 — full revelation	only if there are more messages than types	each type's **incentive compatibility**: no type wants to mimic another
Hybrid (semi-separating)	strictly between prior and 0/1	—	a mixing type must be **indifferent**

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3 · Worked — all four candidates in Gibbons's Fig. 4.2.2 (pp. 188–190)

Two equally likely types; messages L, R; actions u, d. Payoffs (Sender, Receiver):

Type	L, then u	L, then d	R, then u	R, then d
t_1	1, 3	4, 0	2, 1	0, 0
t_2	2, 4	0, 1	1, 0	1, 2

Receiver's beliefs: $p = \mu(t_1 \mid L)$, $q = \mu(t_1 \mid R)$. Two facts used throughout:

$$\text{after } L: \; E[u] = 3p + 4(1-p) = 4 - p \; > \; E[d] = 0 \cdot p + 1 \cdot (1-p) = 1 - p \quad \Rightarrow \; u \text{ for every } p;$$

$$\text{after } R: \; E[u] = 1 \cdot q + 0 = q, \quad E[d] = 0 + 2(1-q) = 2 - 2q \quad \Rightarrow \; d \iff q \leq \tfrac{2}{3}.$$

1. Pooling on L, Sender (L, L). Bayes: $p = 0.5$. Reply to L: u, so t_1 earns 1 and t_2 earns 2. Keep them from R: if the reply to R were u, t_1 would get $2 > 1$ — deviation. So the reply to R must be d (then t_1 gets $0 < 1$, t_2 gets $1 < 2$). d needs $q \leq \frac{2}{3}$.

$$\boxed{[(L, L),\ (u, d),\ p = 0.5,\ q \leq \tfrac{2}{3}] \text{ is a pooling PBE.}}$$

2. Pooling on R, Sender (R, R). Bayes: $q = 0.5 \leq \frac{2}{3}$, so the reply to R is d: t_1 earns 0. But after L the reply is u at **any** p, giving t_1 a payoff of $1 > 0$. **No such PBE.**

3. Separating, $t_1 \to L$, $t_2 \to R$. Bayes: $p = 1$, $q = 0$; replies u after L, d after R; both types earn 1. But t_2 deviating to L is met by u and earns $2 > 1$. **No such PBE.**

4. Separating, $t_1 \to R$, $t_2 \to L$. Bayes: $p = 0$, $q = 1$; replies u after L (always), u after R ($q = 1 > \frac{2}{3}$); both types earn 2. Deviations: t_1 to L gets $1 < 2$; t_2 to R gets $1 < 2$.

$$\boxed{[(R, L),\ (u, u),\ p = 0,\ q = 1] \text{ is a separating PBE.}}$$

> **The method, every time:** (i) fix the Sender's strategy; (ii) Bayes on every used message; (iii) the Receiver's best reply to those beliefs; (iv) for unused messages, pick the reply that best deters deviation and check some belief supports it; (v) check every type's incentive constraint against every other message.

(Whether the pooling PBE in case 1 survives the intuitive criterion is settled in L08-08-intuitive-criterion §4.)

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4 · U — hybrid equilibria (Tadelis Remark, p. 319; Gibbons pp. 186, 202–205)

In a hybrid (semi-separating) equilibrium one type plays a pure message and the other randomizes. If t_1 always sends m and t_2 sends it with probability π:

$$\mu(t_1 \mid m) = \frac{p(t_1)}{p(t_1) + p(t_2)\,\pi} \ \in\ (p(t_1),\, 1) \quad \text{for } \pi \in (0, 1).$$

The message is **informative but not conclusive**. The mixing type must be indifferent, which pins π — worked numerically for Spence in L08-06-spence-job-market §6. Tadelis defers the general treatment to Ch. 17 (reputation), beyond this lecture.

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5 · U — two more applications in Gibbons §4.2 (pp. 205–210)

Corporate investment and capital structure (Myers–Majluf, §4.2.C). An entrepreneur whose existing firm earns $\pi \in \{L, H\}$ ($H > L > 0$, $\Pr\{L\} = p$) needs I to fund a project returning $R > I(1+r)$, and offers an equity stake s. The investor accepts iff $s[qL + (1-q)H + R] \geq I(1+r)$ (4.2.11), with $q = \Pr\{L \mid s\}$; type π prefers to go ahead iff $s \leq R/(\pi + R)$ (4.2.12). Pooling needs

$$\frac{I(1+r)}{pL + (1-p)H + R} \ \leq\ \frac{R}{H + R} \qquad (4.2.13)$$

— the **high type subsidizes the low type** through the larger pooled stake. When (4.2.13) fails there is a separating equilibrium in which the low type is financed at $s = I(1+r)/(L+R)$ and the high type **forgoes a profitable project**: "financing terms that are attractive to the high-profit type are even more attractive to the low-profit type" (p. 207). The signal space (equity stakes) cannot sort, which pushes firms toward debt or internal funds.

Monetary policy (§4.2.D). A two-period version of the policy game, with the central bank's distaste for inflation private: first-period inflation is the message, employers' second-period expectation the reply.

The separating constraint makes the strong type choose **lower** inflation than under complete information — over-signaling, like the over-investment in education in note 06. Named only; the repeated-game background is Lecture 5.

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6 · C — classifying the entry game (Tadelis p. 319)

BNE of L08-01-subgame-perfection-fails	Kind	PBE?
(OO, F)	pooling (both types stay out)	**no** — F is never a best reply
(EO, A)	separating (entry reveals C, staying out reveals W)	**yes**, $\mu(x_3) = 1$

Links

- Prev: L08-03-sequential-equilibrium · Next: L08-05-mba-game (the book's own two-type example, solved with this method).
- The continuous-message version: L08-06-spence-job-market; a continuous-message IO application: L08-07-limit-pricing.
- Rules: rules/07 (signaling; labour-market application) · rules/game-theory item 19 ("signalling needs the asymmetry modelled").
- Adverse selection, the problem signaling responds to: Lecture 7.
- Code: the generic two-type solver `pure_pbe`, tested on this example, in L08-11-python §3.
- Exercises (number and page only): Gibbons Problems 4.3 (p. 246), 4.4 (p. 247), 4.5–4.6 (p. 249); Muñoz-Garcia Ch. 9 Exercises 1 (p. 376), 5 (p. 422), 7 (p. 437); Ch. 11 Exercises 1 (p. 482), 3 (p. 497).

16.1 · The MBA game: one separating and one pooling equilibrium, held up by beliefs

Claim. In Tadelis's MBA game the degree adds nothing to productivity, yet there is a **separating PBE** in which only the high-skill worker studies and is made a manager — because the degree costs him 2 and his low-skill twin 5, and the manager's premium is 4. There is also a **pooling PBE** in which nobody studies and everybody is put in blue-collar work, **held up only by the employer's off-path belief** that an MBA signals low skill with probability at least 5/8. One threshold, $\mu^* = 3/8$, settles every information set.

Prev: L08-04-signaling-games · Next: L08-06-spence-job-market · Map: L08-00-map

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1 · C — the game (pp. 319–321)

1. Nature draws the worker's skill $\theta \in \{H, L\}$, $\Pr\{\theta = H\} = p > 0$; the worker learns it.
2. The worker chooses an MBA (D) or not (U). Private cost $c_H = 2$, $c_L = 5$ (if D), 0 otherwise.
3. The employer sees D or U (not θ) and assigns the worker to manager (M, wage $w_M = 10$) or blue-collar (B, wage $w_B = 6$).
4. Employer's net profit (the degree adds nothing to productivity):

Skill	M	B
H	10	5
L	0	3

Worker's payoff = wage − cost. All eight terminal payoffs (worker, employer), as in Fig. 16.1:

Type, message	M	B
H, U	10, 10	6, 5
H, D	$10-2=8$, 10	$6-2=4$, 5
L, U	10, 0	6, 3
L, D	$10-5=5$, 0	$6-5=1$, 3

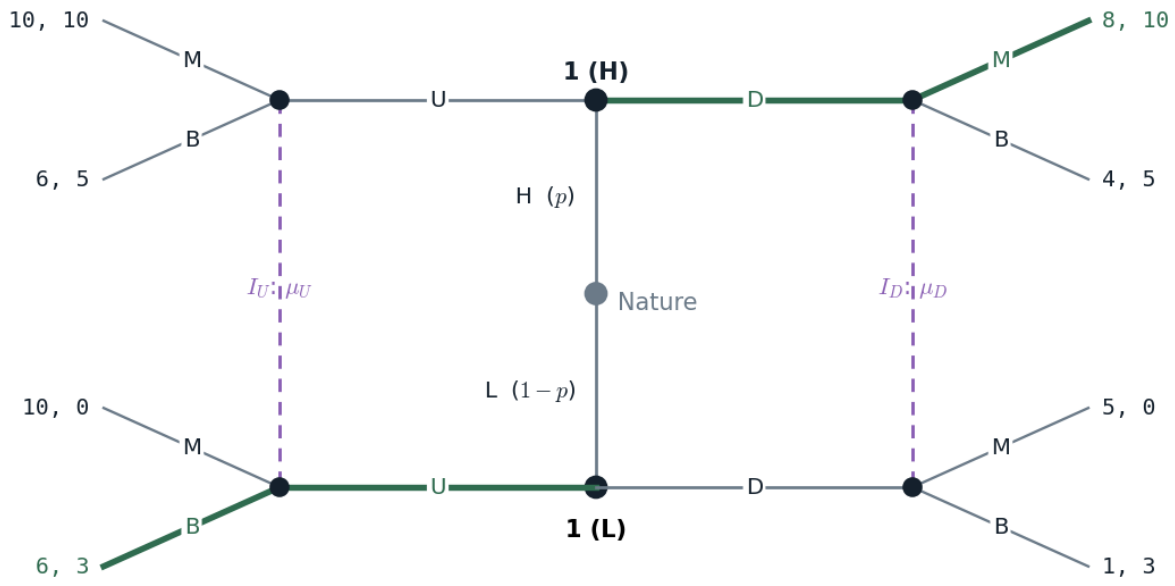


Figure — Tadelis's Fig. 16.1 redrawn with the separating PBE (DU, BM) in green: the high type studies and is made a manager (8, 10); the low type does not and is put in blue-collar work (6, 3). Source: Tadelis pp. 320–322.

The two dashed lines are the employer's information sets I_U and I_D: at each it knows the message, not the type. Only the green branches are played; the employer's replies at both sets are still part of its strategy, which is what lets a reply after an unused message do the work in the pooling equilibrium (§5).

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2 · C — beliefs by Bayes' rule (eqs. 16.1–16.2, p. 321)

Let type H choose U with probability σ_H and type L with σ_L. With $\mu_U = \Pr\{H \mid U\}$ and $\mu_D = \Pr\{H \mid D\}$:

$$\mu_U = \frac{p\,\sigma_H}{p\,\sigma_H + (1-p)\,\sigma_L} \quad (16.1), \qquad \mu_D = \frac{p\,(1-\sigma_H)}{p\,(1-\sigma_H) + (1-p)(1-\sigma_L)} \quad (16.2).$$

- $\sigma_H = \sigma_L = 1$ (both choose U): $\mu_U = p$; (16.2) is $\frac{0}{0}$ — μ_D is **free**.
- $\sigma_H = \sigma_L = 0$ (both choose D): $\mu_D = p$; μ_U free.
- $\sigma_H = 0,\ \sigma_L = 1$ (strategy DU): $\mu_U = \dfrac{0}{0 + (1-p)} = 0$, $\mu_D = \dfrac{p}{p+0} = 1$ (footnote 2, p. 322).

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3 · C — the employer's best reply at any information set (p. 323)

At belief μ:

$$E[\pi(M)] = 10\mu + 0\,(1-\mu) = 10\mu, \qquad E[\pi(B)] = 5\mu + 3(1-\mu) = 3 + 2\mu.$$

$$M \text{ optimal} \iff 10\mu \geq 3 + 2\mu \iff 8\mu \geq 3 \iff \boxed{\mu \geq \mu^* = \tfrac{3}{8}}.$$

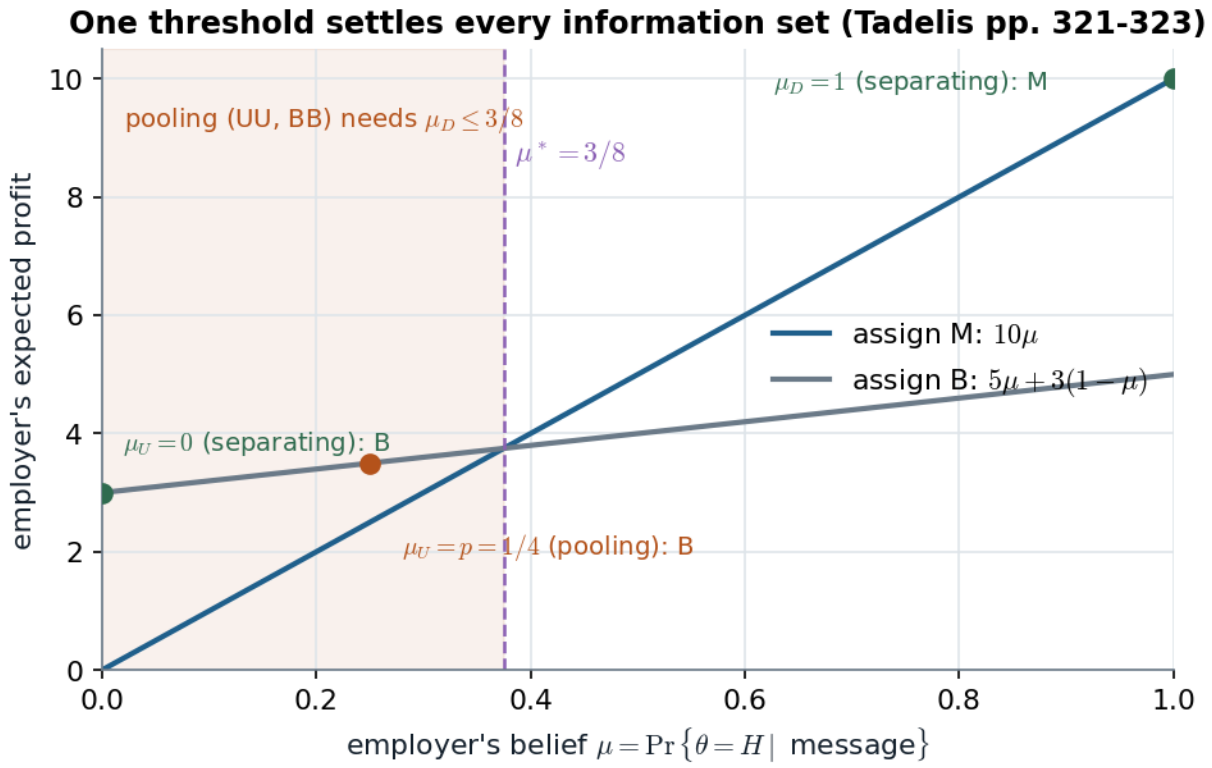


Figure — one threshold settles every information set: M iff $\mu \geq 3/8$. Separating: $\mu_U = 0 \Rightarrow B$, $\mu_D = 1 \Rightarrow M$. Pooling on U: $\mu_U = p = 1/4 < 3/8 \Rightarrow B$, and the unused D must be met with a belief in the shaded band. Source: Tadelis pp. 321–323.

Everything the employer does in any equilibrium is a reading of this one picture: where the belief at I_U and at I_D falls relative to 3/8. The prior $p = 1/4$ sits left of the threshold — which is why pooling on U gets B, not M, at the book's numbers (§6 asks what happens when it does not).

---

4 · C — the Bayesian normal form at $p = 1/4$ (p. 322)

Worker strategies $a_H a_L$, employer strategies $a_U a_D$. One cell in full: (UD, MB) — with prob. $\frac{1}{4}$ type H plays U, gets M: $(10, 10)$; with prob. $\frac{3}{4}$ type L plays D, gets B: $(6-5, 3) = (1, 3)$:

$$(v_1, v_2) = \tfrac{1}{4}(10, 10) + \tfrac{3}{4}(1, 3) = (2.5 + 0.75,\ 2.5 + 2.25) = (3.25,\ 4.75).$$

	MM	MB	BM	BB
UU	10, 2.5	10, 2.5	6, 3.5	6, 3.5
UD	6.25, 2.5	3.25, 4.75	5.25, 1.25	2.25, 3.5
DU	9.5, 2.5	8.5, 1.25	6.5, 4.75	$5.5^{\dagger}$, 3.5
DD	5.75, 2.5	1.75, 3.5	5.75, 2.5	1.75, 3.5

† (DU, BB): $\frac{1}{4}(6-2)+\frac{3}{4}(6) = 1+4.5 = 5.5$; the book prints 4.5. It changes no best response (column BB's maximum is UU's 6 either way).

Best responses. Worker, by column: $MM \to UU$ (10); $MB \to UU$ (10); $BM \to DU$ (6.5); $BB \to UU$ (6). Employer, by row: $UU \to BM$ or BB (3.5); $UD \to MB$ (4.75); $DU \to BM$ (4.75); $DD \to MB$ or BB (3.5). Mutual best responses:

$$\boxed{(UU, BB) \quad \text{and} \quad (DU, BM)} \qquad \text{— the two pure BNE (p. 322).}$$

((UU, BM) fails: against BM the worker prefers DU, $6.5 > 6$.)

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5 · Worked — which BNE are PBE (pp. 322–323)

5.1 · (DU, BM) — separating

Both messages are used, so by **Prop. 15.1** the BNE with Bayes beliefs $\mu_U = 0$, $\mu_D = 1$ is a PBE. Checking every information set anyway:

Who	Belief	Prescribed	Payoff	Deviation	Payoff	Holds because
Employer at I_U	$\mu_U = 0 < \frac{3}{8}$	B	3	M	0	$3 > 0$
Employer at I_D	$\mu_D = 1 \geq \frac{3}{8}$	M	10	B	5	$10 > 5$
Worker H	—	D	$10-2=8$	U $(\to B)$	6	$c_H = 2 < w_M - w_B = 4$
Worker L	—	U	6	D $(\to M)$	$10-5=5$	$c_L = 5 > w_M - w_B = 4$

The two incentive-compatibility constraints are the whole story:

$$c_H \;\leq\; w_M - w_B \;\leq\; c_L \qquad (2 \leq 4 \leq 5).$$

The premium must be worth the degree to the high type and not worth it to the low type. This is the discrete version of Spence's single-crossing condition (note 06).

5.2 · (UU, BB) — pooling

I_U is on the path: $\mu_U = p = \frac{1}{4} < \frac{3}{8}$, so B is a best reply. I_D is **off** the path (nobody studies): μ_D is free, but B there must be a best reply:

$$5\mu_D + 3(1-\mu_D) \;\geq\; 10\mu_D \iff 3 \geq 8\mu_D \iff \mu_D \leq \tfrac{3}{8}.$$

Worker: U gives 6 to both types; D (met with B) gives $6-2=4$ to H and $6-5=1$ to L. Nobody deviates.

$$\boxed{(UU, BB),\ \mu_U = \tfrac{1}{4},\ \mu_D \in [0, \tfrac{3}{8}] \text{ is a pooling PBE.}}$$

The pooling equilibrium lives on a belief: if an MBA were read as "high skill with probability above 3/8", the employer would promote, and the high type would study ($8 > 6$). The belief that sustains it — *an MBA means probably low skill* — is exactly what the intuitive criterion calls unreasonable (L08-08-intuitive-criterion §2).

PBE	Kind	Beliefs	Worker payoffs (H, L)	Employer (expected)
(DU, BM)	separating	$\mu_U = 0,\ \mu_D = 1$	8, 6	$\frac{1}{4} \cdot 10 + \frac{3}{4} \cdot 3 = 4.75$
(UU, BB)	pooling	$\mu_U = \frac{1}{4},\ \mu_D \leq \frac{3}{8}$	6, 6	$\frac{1}{4} \cdot 5 + \frac{3}{4} \cdot 3 = 3.5$

---

6 · Worked extension — what the prior changes (same model, derived here)

The book fixes $p = \frac{1}{4}$. Varying p with everything else at the book's values:

- **Separating** (DU, BM) uses beliefs 0 and 1 and the constraints $c_H \leq 4 \leq c_L$: **it does not depend on p at all.**
- **Pooling on** U gets the reply the prior dictates: B if $p < \frac{3}{8}$, M if $p > \frac{3}{8}$, both at $p = \frac{3}{8}$. With M after U, both types earn 10 and no one gains from D (at most $10 - 2 = 8$), so (UU, MM) and (UU, MB) are PBE for any μ_D that supports the reply after D.
- **Pooling on** D never works at the book's numbers: after D the reply is M or B; if M, type L earns 5 and would switch to U unless U is met with B **and** $6 < 5$ — impossible; if B, both earn less than the 6 that U guarantees.

p	Pure PBE (book's costs and wages)
$p < \frac{3}{8}$ (e.g. $\frac{1}{4}$)	(DU, BM); (UU, BB) with $\mu_D \leq \frac{3}{8}$
$p = \frac{3}{8}$	(DU, BM); (UU, BB); (UU, MM); (UU, MB)
$p > \frac{3}{8}$	(DU, BM); (UU, MM) with $\mu_D \geq \frac{3}{8}$; (UU, MB) with $\mu_D \leq \frac{3}{8}$

Interactive companion. The MBA Game (`companions/companion-mba-game.html`) — move the prior across 3/8 and watch pooling on U switch from B to M; push c_L below $w_M - w_B$ and the separating equilibrium disappears (the low type would mimic); the table lists every pure PBE, its supporting beliefs and the intuitive-criterion verdict.

Links

- Prev: L08-04-signaling-games · Next: L08-06-spence-job-market.
- The refinement that removes (UU, BB): L08-08-intuitive-criterion §2.
- Prop. 15.1 and the requirements used above: L08-02-beliefs-and-pbe §4–6.
- Rules: rules/07 (labour-market signaling).
- Code: the 16-cell matrix, the BNE and the PBE enumeration, all asserted, in L08-11-python §4.
- Exercises (number and page only): Tadelis 16.4 "School Choice" (p. 336), 16.5 "Drug Ads" (p. 337); Muñoz-Garcia Ch. 9 Exercises 2 (p. 385), 3 (p. 394), 4 (p. 406); MWG 13.C.5 (pp. 474–475).

4.2.B · Spence's job market: how much education it takes to be believed

Claim. When education costs the able less **at the margin** (single crossing), a high-ability worker can separate by acquiring education the low type will not copy. Competitive firms then pay each type its productivity, the low type gets **no** education, and the high type gets **more than he would under full information** — at least enough that mimicking costs the low type the whole wage premium. With M&R's numbers: **separation needs** $e_H \in [1.5, 3]$, the least-cost level is **1.5**, and the education is **pure social waste**. Whether the high type gains from being *able* to signal depends on the prior: at 30% high types he does ($35 > 29$); above 50% he would be better off if signaling were impossible.

Prev: L08-05-mba-game · Next: L08-07-limit-pricing · Map: L08-00-map

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1 · C — the model (Gibbons pp. 190–194)

1. Nature draws ability $\eta \in \{H, L\}$, $\Pr\{\eta = H\} = q$ (this note writes p, as in Tadelis; MWG writes λ).
2. The worker learns η and chooses education $e \geq 0$.
3. Two firms observe e (not η) and simultaneously offer wages; the worker takes the higher.

Payoffs: worker $w - c(\eta, e)$; employing firm $y(\eta, e) - w$; the other firm 0. High ability is more productive, $y(H, e) > y(L, e)$, and education does not reduce output, $y_e \geq 0$ (footnote 5). Spence's (1973) stark case: $y(\eta, e) = y(\eta)$ — **education is unproductive.**

The crucial assumption — single crossing (p. 192):

$$\boxed{c_e(L, e) > c_e(H, e) \quad \text{for every } e}$$

The **marginal** cost of education is higher for the low type, so his indifference curves in the (e, w) plane are **steeper** (Fig. 4.2.3): to stay as well off after one more unit of education he needs a larger raise.

Competition pins the wage to the belief (4.2.1, p. 193). With firms sharing a belief $\mu(H \mid e)$ after any e (on or off the path), Bertrand-style competition drives expected profit to zero:

$$w(e) = \mu(H \mid e)\, y(H, e) + [1 - \mu(H \mid e)]\, y(L, e). \qquad (4.2.1)$$

(4.2.1) replaces Signaling Requirement 2R (L08-04-signaling-games §2) in this two-Receiver game.

The benchmark. Under complete information $w = y(\eta, e)$ and type η chooses $e^*(\eta) = \arg\max_e y(\eta, e) - c(\eta, e)$, earning $w^*(\eta) = y[\eta, e^*(\eta)]$ (Fig. 4.2.4). With unproductive education, $e^*(\eta) = 0$.

No envy vs envy (pp. 194–196). If $w^*(L) - c[L, e^*(L)] > w^*(H) - c[L, e^*(H)]$ the low type would not copy the high type even if copying earned him $w^*(H)$ (Fig. 4.2.5): the full-information outcome separates by itself. The interesting case — and the only one with unproductive education — is **envy** (Fig. 4.2.6): the low type *would* copy, so the high type must do more.

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2 · C — separating equilibria (Gibbons pp. 199–202; MWG Lemmas 13.C.1–13.C.2, pp. 453–454)

In a separating PBE beliefs are 0 or 1 on the path, so by (4.2.1) each type is paid its own productivity (**MWG Lemma 13.C.1**). The low type is then paid $y(L,e)$ whatever he does on the path, so he picks his full-information education: $e(L) = e^*(L)$ — with unproductive education, **zero** (**MWG Lemma 13.C.2**: any $e > 0$ would cost him and earn nothing).

The high type picks e_s such that the low type will not mimic:

$$\underbrace{w^*(L) - c[L, e^*(L)]}_{\text{L's separating payoff}} \;\geq\; \underbrace{y(H, e_s) - c(L, e_s)}_{\text{L's payoff from mimicking}} .$$

Off-path beliefs that support it (p. 200): **"high iff $e \geq e_s$"**, giving $w(e) = y(L,e)$ for $e < e_s$ and $y(H,e)$ for $e \geq e_s$. Under this schedule the low type's best $e < e_s$ is $e^*(L)$; for $e \geq e_s$ his best is e_s (steeper curves), and he is indifferent at the least-cost level (resolved in favour of $e^*(L)$). The high type's best is e_s, because his flatter indifference curve through $(e_s, y(H, e_s))$ lies above $y(L,e)$ for every $e < e_s$.

> **The high type over-invests:** $e_s > e^*(H)$ in the envy case — "the high-ability worker must choose $e_s > e^*(H)$ … because the low-ability worker will mimic any value of e between $e^*(H)$ and e_s" (p. 200). **Any** $e \geq e_s$ small enough that the high type still prefers it to being thought low is also a separating PBE (p. 202): a **continuum**.

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3 · Worked — Milgrom & Roberts's numbers (pp. 155–156)

High productivity $\theta_H = \$50$/hour, low $\theta_L = \$20$; 30% high; education unproductive; cost per unit of education $c_H = \$10$, $c_L = \$20$ (so $c(\theta, e) = c_\theta\, e$, linear, and single crossing is $c_H < c_L$). The low type chooses $E_L = 0$. M&R's two **self-selection constraints**:

$$\text{L must not mimic:}\quad 50 - c_L E_H \leq 20 \iff 50 - 20E_H \leq 20 \iff E_H \geq \frac{50-20}{20} = 1.5,$$

$$\text{H must prefer to signal:}\quad 50 - c_H E_H \geq 20 \iff 50 - 10E_H \geq 20 \iff E_H \leq \frac{50-20}{10} = 3.$$

$$\boxed{E_H \in \left[\tfrac{\theta_H - \theta_L}{c_L},\ \tfrac{\theta_H - \theta_L}{c_H}\right] = [1.5,\ 3]}$$

M&R's check with $E_H = 2$ (p. 156): mimicking nets the low type $50 - 20 \times 2 = \$10 < \$20$; the high type nets $50 - 10 \times 2 = \$30 > \$20$. Both constraints hold. The least-cost choice is "a little bit more than 1.5".

> **"If $C_H \leq C_L$" fails — i.e. if $c_H \geq c_L$ — "these inequalities cannot both be satisfied" (p. 156):** the interval is empty, because any education the high type would buy to be believed the low type buys too. **The signal works only because it hurts the low type more.**

Pooling wage. With no signal, both types earn expected productivity $(0.3 \times 50) + (0.7 \times 20) = \$29$ (p. 155).

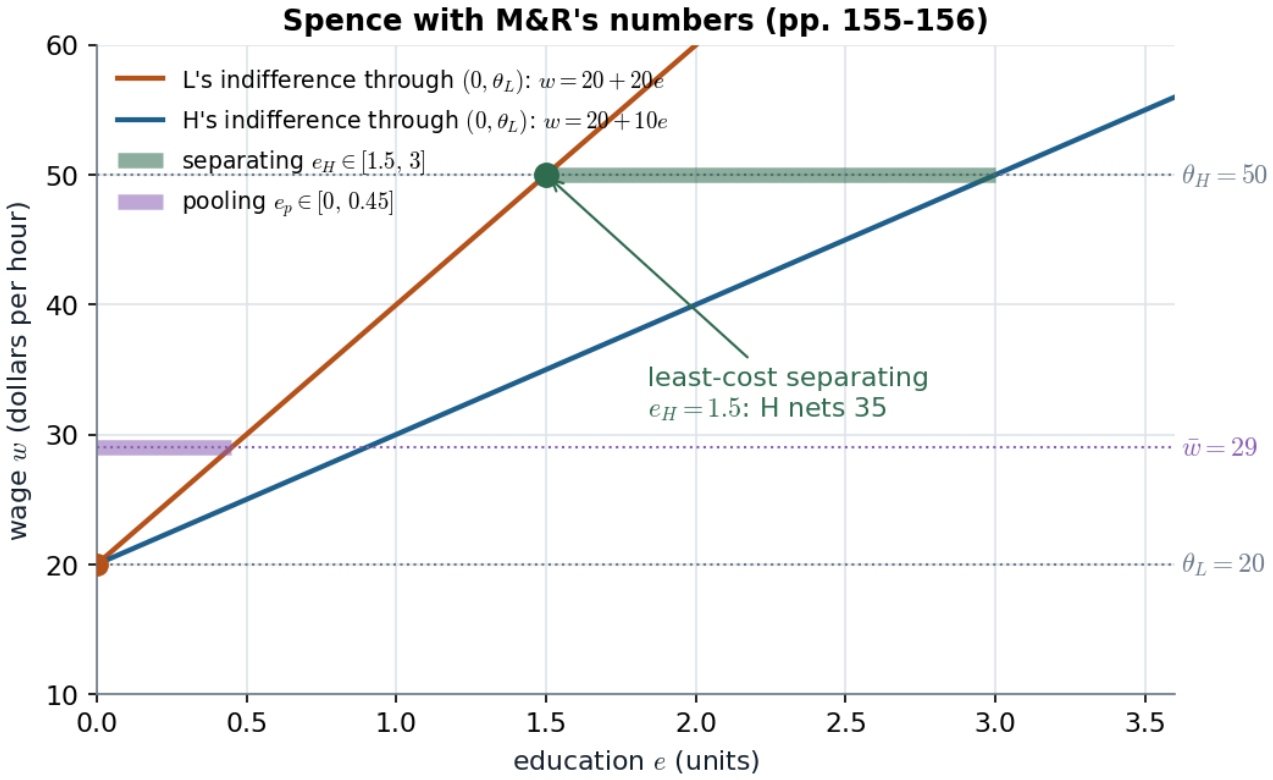


Figure — the two indifference lines through the low type's separating point $(0, \theta_L)$ cut the θ_H line at $e = 1.5$ (L indifferent to mimicking) and $e = 3$ (H indifferent to signaling): the separating set. The pooled wage line carries the pooling set $[0, 0.45]$ of §4. Source: M&R pp. 155–156; Gibbons Fig. 4.2.8.

Linear costs make the indifference curves straight lines, so the picture is exact: the steeper (orange) line is the low type's, the flatter (blue) one the high type's — single crossing is simply "orange steeper than blue". Every e_H on the green segment is a separating PBE; the dot at 1.5 is the cheapest one.

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4 · C — pooling equilibria (Gibbons pp. 196–199)

Both types choose e_p; Bayes gives $\mu(H \mid e_p) = q$, so

$$w_p = q\, y(H, e_p) + (1 - q)\, y(L, e_p). \qquad (4.2.2)$$

The harshest supporting off-path belief is "**low** for any $e \neq e_p$" (4.2.3), so $w(e) = y(L, e)$ off the path (4.2.4), and each type compares e_p with its best e against $y(L, e)$, i.e. $e^*(L)$ (4.2.5). Gibbons stresses (p. 197): *"Although this belief may seem strange, nothing in the definition of perfect Bayesian equilibrium rules it out."*

With M&R's numbers ($w_p = 29$, off-path wage 20 at $e = 0$): type θ pools iff $29 - c_\theta e_p \geq 20$:

$$\text{L: } e_p \leq \frac{9}{20} = 0.45, \qquad \text{H: } e_p \leq \frac{9}{10} = 0.9 \qquad \Rightarrow \qquad \boxed{e_p \in [0,\ 0.45]}$$

— in general $e_p \leq q(\theta_H - \theta_L)/\max(c_H, c_L)$. Another continuum, and pooling at $e_p > 0$ burns education for no information at all.

5 · C — who gains from signaling (MWG pp. 456–457, Fig. 13.C.8; M&R p. 156)

Compare the least-cost separating outcome with a world where signaling is impossible (everyone paid w_p at $e = 0$):

	Least-cost separating	No signaling	Verdict
Low type	$\theta_L = 20$	29	**always worse off** ($w_p > \theta_L$ whenever $q > 0$)
High type	$\theta_H - c_H \dfrac{\theta_H - \theta_L}{c_L} = 50 - 10(1.5) = 35$	29	better off **here**; in general iff $q < q^*$
Firms	zero profit	zero profit	indifferent

The high type gains iff

$$\theta_H - \frac{c_H}{c_L}(\theta_H - \theta_L) \;\geq\; q\theta_H + (1-q)\theta_L \iff (1-q)(\theta_H - \theta_L) \geq \frac{c_H}{c_L}(\theta_H - \theta_L) \iff \boxed{q \leq q^* = 1 - \frac{c_H}{c_L}} = 1 - \tfrac{10}{20} = 0.5.$$

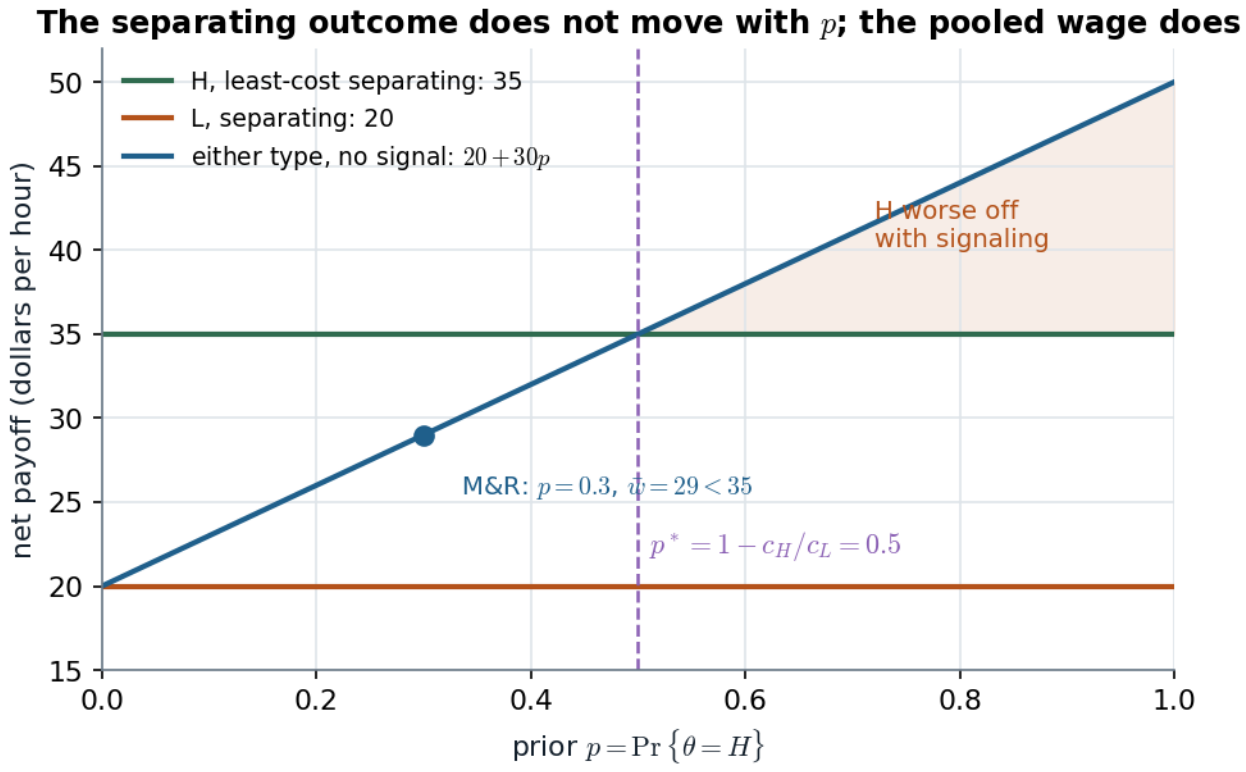


Figure — the separating outcome is the same at every prior (it never uses p), while the pooled wage rises with p; above $p^* = 1 - c_H/c_L = 0.5$ the high type would rather live where signaling is impossible. Source: MWG p. 456, Fig. 13.C.8; M&R pp. 155–156 numbers.

The flat green line is the striking fact: **the separating equilibrium does not depend on the prior at all** (MWG: "the set of separating equilibria is completely unaffected by the fraction λ"). As high types become common, the pooled wage approaches θ_H and the education they burn to avoid being mistaken for "the handful of bad workers" becomes pure loss.

> **Efficiency (M&R p. 156).** Education here has no productive value, so *"the expenditure on education in this example is a pure social waste!"* The expected waste per worker in the least-cost separating equilibrium is $q\, c_H\, e_s = 0.3 \times 10 \times 1.5 = \$4.50$ per hour.
>
> **Interactive companion.** The Education Signal (`companions/companion-spence-signaling.html`) — move θ_L, θ_H, the prior and both cost rates; watch the separating interval, the least-cost e^*, the pooling interval and p^*; push c_H up to c_L and separation vanishes.

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6 · U/C — hybrid equilibria (Gibbons pp. 202–205)

The high type chooses e_h; the low type randomizes: e_h with prob. π, $e^*(L)$ with $1-\pi$.

Belief after e_h (4.2.8), by Bayes:

$$\mu(H \mid e_h) = \frac{q}{q + (1-q)\pi} \equiv r.$$

Low type indifferent (4.2.9): $w^*(L) - c[L, e^*(L)] = w_h - c(L, e_h)$. **Wage** (4.2.10): $w_h = r\, y(H, e_h) + (1-r)\, y(L, e_h)$. Solve (4.2.10) for r, then (4.2.8) for π:

$$\pi = \frac{q(1-r)}{r(1-q)}.$$

With M&R's numbers and $e_h = 1$: indifference gives $w_h = 20 + 20 \times 1 = 40$; then $40 = 50r + 20(1-r) \Rightarrow r = \frac{20}{30} = \frac{2}{3}$; and

$$\pi = \frac{0.3 \times \frac{1}{3}}{\frac{2}{3} \times 0.7} = \frac{0.1}{0.4667} = 0.214.$$

A hybrid needs $q \le r \le 1$: $r \ge q$ gives $e_h \ge q(\theta_H - \theta_L)/c_L = 0.45$ (the pooling bound, $\pi \to 1$), and $r \le 1$ gives $e_h \le 1.5$ (the least-cost separating level, $\pi \to 0$). **Hybrids connect the pooling and separating families**: "the separating equilibrium … is the limit of the hybrid equilibria" (p. 204).

e_h	w_h	r	π
0.45	29	0.30	1 (pooling)
1.00	40	0.667	0.214
1.50	50	1	0 (separating)

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7 · G — MWG §13.C in one paragraph (pp. 450–459)

MWG's version: $\theta_H > \theta_L > 0$, $\lambda = \Pr\{\theta_H\}$, cost $c(e,\theta)$ with $c_e > 0$, $c_{ee} > 0$, $c_\theta < 0$, $c_{e\theta} < 0$; utility $w - c(e,\theta)$; PBE with the firms' common belief $\mu(e)$ (equivalent to sequential equilibrium here). Lemma 13.C.1: separating wages equal productivities; Lemma 13.C.2: $e^*(\theta_L) = 0$; a continuum of separating and of pooling equilibria follows, exactly as in §2–§4 above; signaling can Pareto-dominate or be Pareto-dominated by no signaling (Fig. 13.C.8). The refinement that selects $\tilde{e}$ is in L08-08-intuitive-criterion §5. MWG Exercise 13.C.4 (continuum of types, $c = e^2/\theta$) is beyond the two-type case. Not examinable unless the lecture brings it.

Links

- Prev: L08-05-mba-game · Next: L08-07-limit-pricing.
- The refinement that selects $e_H = 1.5$: L08-08-intuitive-criterion §5.
- Adverse selection without a signal (the lemons logic the signal escapes): Lecture 7.
- Rules: rules/07 §Signaling, §Application: labour-market signaling.
- Code: `spence` and `hybrid`, asserted on M&R's numbers, in L08-11-python §6.
- Practice already in the project: `Listas/pratica-aula-01-02.md` item 18 (M&R Food for Thought 5.2, p. 164, signals of intent) with its solution `Listas/pratica-aula-01-02-solutions/item18.tex`.
- Exercises (number and page only): M&R Ch. 5 Quantitative Q 5.2 (p. 164, extends this example); Gibbons Problem 4.7 (p. 249); MWG 13.C.1, 13.C.2, 13.C.4 (pp. 474–475); Muñoz-Garcia Ch. 11 Exercise 4 (p. 500).

16.2 · Limit pricing: an incumbent over-produces to prove it is cheap

Claim. Bain's limit pricing — charge a low price now to keep entrants out — survives a formal model (Milgrom–Roberts 1982, simplified by Tadelis) **as a signal**: a low-cost incumbent produces **more than its monopoly output** in period 1 so that a high-cost incumbent would not find it worth copying. Separating PBE need $q_1^L \in [2.62, 3.32]$ (monopoly output is 2); pooling PBE need $q_1 \in [1.49, 1.92]$. Both are continua — until the intuitive criterion (note 08) leaves only $q_1^L = 2.62$.

Prev: L08-06-spence-job-market · Next: L08-08-intuitive-criterion · Map: L08-00-map

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1 · C — the game (pp. 323–324)

Two periods, inverse demand $P = 5 - q$ each period. The incumbent (firm 1) has marginal cost $c_1 \in \{1, 2\}$ — **L**ow or **H**igh — each with probability $\frac{1}{2}$, known only to itself. The entrant (firm 2) has $c_2 = 2$ and a fixed entry cost $F = \frac{1}{2}$.

1. Nature draws c_1.
2. Firm 1 produces q_1^1 as a monopolist; price $5 - q_1^1$.
3. Firm 2 observes q_1^1 (not c_1) and enters or not.
4. No entry: firm 1 is a monopolist again in period 2. Entry: Cournot duopoly, $P = 5 - q_1^2 - q_2^2$.

Notation: $q_1^{t\theta}$ is type θ's output in period t; $\mu(q_1^1) = \Pr\{c_1 = 2 \mid q_1^1\}$ is the entrant's belief **that the incumbent is high-cost**; "no entry" is coded $q_2^2 = 0$ (footnote 3). No discounting.

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2 · C — the building blocks (Steps 1–2, pp. 324–325)

Step 1 — monopoly. $\max_q (5 - q - c_1)q$: FOC $5 - 2q - c_1 = 0 \Rightarrow q = \dfrac{5 - c_1}{2}$.

Type	Monopoly q	Price	Monopoly profit
L ($c_1 = 1$)	2	3	$(3-1)\cdot 2 = 4$
H ($c_1 = 2$)	1.5	3.5	$(3.5-2)\cdot 1.5 = 2.25$

Step 2 — Cournot with known cost. FOCs $5 - 2q_1 - q_2 - c_1 = 0$ and $5 - q_1 - 2q_2 - 2 = 0$ give $BR_1(q_2) = \dfrac{5 - q_2 - c_1}{2}$ and $BR_2(q_1) = \dfrac{3 - q_1}{2}$.

- **Known H** ($c_1 = 2$): $q_1 = \dfrac{3 - q_2}{2}$, $q_2 = \dfrac{3 - q_1}{2} \Rightarrow q_1 = q_2 = 1$, $P = 3$; $v_1 = (3-2)\cdot 1 = 1$, $v_2 = (3-2)\cdot 1 - \frac{1}{2} = \frac{1}{2} > 0$ — **entry pays**.
- **Known L** ($c_1 = 1$): $q_1 = \dfrac{4 - q_2}{2}$, $q_2 = \dfrac{3 - q_1}{2} \Rightarrow$ substitute: $q_1 = \dfrac{4 - (3 - q_1)/2}{2} \Rightarrow 4q_1 = 8 - 3 + q_1 \Rightarrow q_1 = \frac{5}{3}$, $q_2 = \frac{2}{3}$, $P = \frac{8}{3}$; $v_1 = (\frac{8}{3} - 1)\frac{5}{3} = \frac{25}{9}$, $v_2 = (\frac{8}{3} - 2)\frac{2}{3} - \frac{1}{2} = \frac{4}{9} - \frac{1}{2} = -\frac{1}{18} < 0$ — **entry does not pay**.

  The entrant wants in only against the high-cost type. So a low-cost incumbent wants to be *recognized*, and a high-cost incumbent wants to be *mistaken* for low cost.

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3 · C — separating equilibria (§16.2.1, pp. 325–330)

Claim 16.1. In any separating PBE, after the L type's output the entrant believes $c_1 = 1$, stays out, and L produces $q_1^{2L} = 2$; after the H type's output it believes $c_1 = 2$, enters, and both produce $q_1^{2H} = q_2^2 = 1$.

Claim 16.2. In any separating PBE, $q_1^{1H} = 1.5$ (H's monopoly output). *Proof:* H is going to face entry and Cournot profit 1 anyway. If it produced anything but 1.5 in period 1, switching to 1.5 would raise period-1 profit and could not lower period-2 profit — after the switch the entrant either keeps believing $c_1 = 2$ (same Cournot game) or believes less, which only helps firm 1. ■

Step 4 — off-path beliefs. Make every deviation as unattractive as possible: believe **high cost** after any output other than q_1^{1L} — $\mu(q_1^1) = 0$ if $q_1^1 = q_1^{1L}$, and 1 otherwise — so the entrant enters (with $q_2^2 = 1$) after anything but q_1^{1L}.

Step 5 — what can q_1^{1L} be? Two incentive-compatibility constraints.

First try: L's monopoly output, $q_1^{1L} = 2$. H's equilibrium payoff: $2.25 + 1 = 3.25$. H mimicking 2: period 1 $(5 - 2 - 2) \cdot 2 = 2$, then no entry and monopoly 2.25:

$$2 + 2.25 = 4.25 > 3.25 \qquad \text{(p. 328)} \text{ — H mimics, so } q_1^{1L} = 2 \text{ is not separating.}$$

The sacrifice in period 1 ($2.25 \to 2$) is small next to the gain in period 2 ($1 \to 2.25$).

IC of H (H must not mimic) — lower bound on q_1^{1L}:

$$(5 - q - 2)q + 2.25 \;\le\; 3.25 \iff (3 - q)q \le 1 \iff q^2 - 3q + 1 \ge 0 \iff q \;\ge\; \frac{3 + \sqrt{5}}{2} = 2.618.$$

(The other branch, $q \le \frac{3-\sqrt{5}}{2} = 0.382$, also stops H from mimicking, but it violates L's own constraint below, which needs $q \ge 0.677$.)

IC of L (L must not deviate) — upper bound. Any deviation triggers entry with $q_2^2 = 1$; L's best deviation is its monopoly output 2 (profit 4) followed by the best reply to $q_2^2 = 1$: $\max_q (5 - q - 1 - 1)q$, at $q = 1.5$, profit 2.25. Best deviation total: $4 + 2.25 = 6.25$. So

$$(5{-}q{-}1)q{+}4 \;\ge\; 6.25 \iff (4{-}q)q \ge 2.25 \iff q^2{-}4q{+}2.25 \le 0 \iff q \in \left[2{-}\sqrt{1.75},\; 2{+}\sqrt{1.75}\right] = [0.677,\; 3.323].$$

Conclusion (p. 329). A continuum of separating PBE: $q_1^{1H} = 1.5$, $q_1^{2L} = 2$, $q_1^{2H} = 1$, beliefs and entry as in Step 4, and

$$\boxed{2.62 \;\le\; q_1^{1L} \;\le\; 3.32}$$

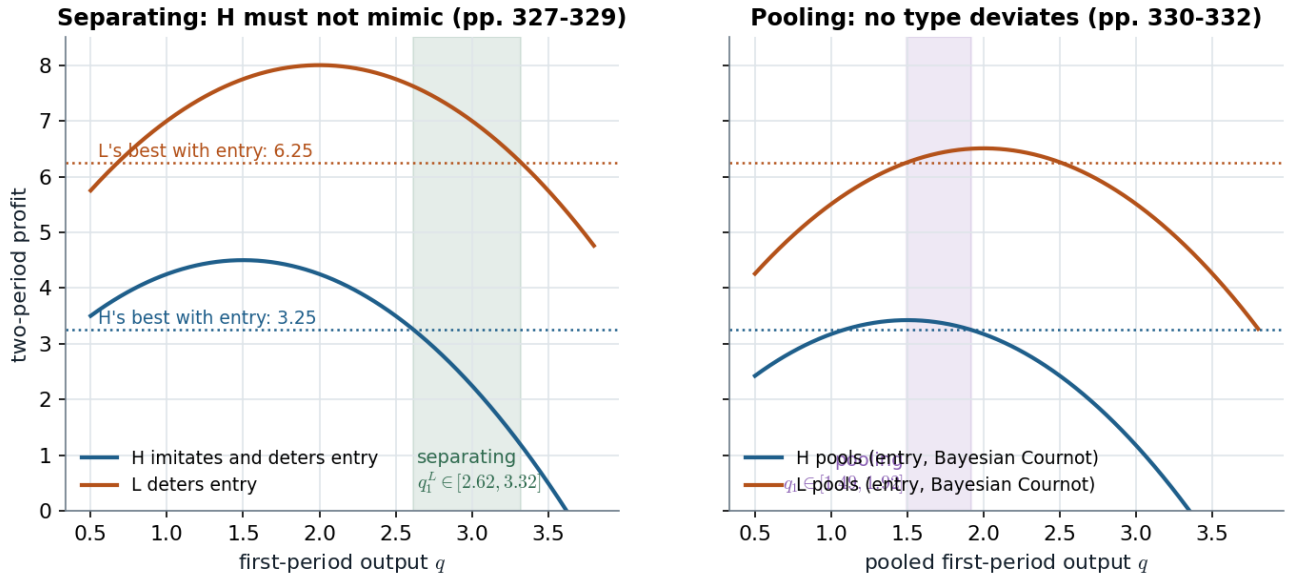


Figure — every incentive constraint of §16.2 as a parabola against a flat line: separating needs H's mimicking payoff below 3.25 and L's payoff above 6.25 (left); pooling needs both types above their best deviation (right). Source: Tadelis pp. 327–332; all functions exact.

Read the left panel from the right edge leftward: L would happily produce as much as 3.32 to be believed, but H would mimic anything below 2.62 — the band in between is the separating set. The right panel's band is narrower because both types must stay put at the *same* output.

> **The signal is credible because it is expensive for the type that would fake it** (p. 329): "for a signal to be credible, it must be the case that no other type would want to use the signal." Here L's price ($5 - 2.62 = 2.38$) is still above its marginal cost 1; with a growing market or more periods of deterrence the required output rises until the price falls **below** marginal cost — Bain's predatory limit pricing (pp. 329–330). Discounting the second period can change the first-try verdict (footnote 4; Tadelis Exercise 16.2, p. 335).

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4 · C — pooling equilibria (§16.2.2, pp. 330–332)

Both types produce the same $\bar{q}_1$, so the entrant learns nothing: $\mu(\bar{q}_1) = \frac{1}{2}$. If it enters, period 2 is a **Cournot game with incomplete information** (Lecture 7). Three best replies:

$$q_1^{2L} = \frac{4 - q_2^2}{2}, \qquad q_1^{2H} = \frac{3 - q_2^2}{2}, \qquad q_2^2 = \frac{3 - \frac{1}{2}(q_1^{2L} + q_1^{2H})}{2}.$$

Solve: $q_1^{2L} + q_1^{2H} = \dfrac{7 - 2q_2^2}{2}$, so $q_2^2 = \dfrac{3 - \frac{7 - 2q_2^2}{4}}{2} \Rightarrow 8q_2^2 = 12 - 7 + 2q_2^2 \Rightarrow q_2^2 = \frac{5}{6}$; then $q_1^{2L} = \dfrac{4 - 5/6}{2} = \frac{19}{12}$ and $q_1^{2H} = \dfrac{3 - 5/6}{2} = \frac{13}{12}$.

Profits: H $= (5 - \frac{13}{12} - \frac{5}{6} - 2)\frac{13}{12} = (\frac{13}{12})^2 = 1.174$; L $= (\frac{19}{12})^2 = 2.507$; entrant $= \frac{5}{6}(3 - \frac{5}{6} - \frac{16}{12}) - \frac{1}{2} = \frac{5}{6} \cdot \frac{5}{6} - \frac{1}{2} = \frac{25}{36} - \frac{18}{36} = \frac{7}{36} > 0$. *(The book prints $\frac{7}{18}$ on p. 331; the arithmetic gives $\frac{7}{36}$. Either way it is positive, so the entrant enters after the pooled output.)*

Off the path: believe **high cost** ($\mu = 1$) after any $q_1^1 \neq \bar{q}_1$, so deviations meet entry with $q_2^2 = 1$; best deviations are worth 3.25 (H) and 6.25 (L), as before.

IC of H: $(3 - \bar{q})\bar{q} + \left(\frac{13}{12}\right)^2 \geq 3.25 \iff \bar{q}^2 - 3\bar{q} + \frac{299}{144} \leq 0$. Discriminant $9 - \frac{1196}{144} = \frac{100}{144}$, root $\frac{10}{12}$: $\bar{q} \in [\frac{3-5/6}{2}, \frac{3+5/6}{2}] = [1.083,\ 1.917]$. (ICH)

IC of L: $(4-\bar{q})\bar{q}+\left(\frac{19}{12}\right)^2 \geq 6.25 \iff \bar{q}^2-4\bar{q}+\frac{539}{144} \leq 0$. Discriminant $16-\frac{2156}{144} = \frac{148}{144}$, root $\frac{\sqrt{148}}{12} = 1.014$: $\bar{q} \in [1.493,\ 2.507]$. (ICL)

> **Conclusion (p. 332).** A continuum of pooling PBE: $q_1^{2L} = \frac{19}{12}$, $q_1^{2H} = \frac{13}{12}$, $q_2^2 = \frac{5}{6}$ after $\bar{q}_1$ and 1 otherwise, beliefs $\frac{1}{2}$ on the path and 1 off it, and
>
> $$\boxed{1.493 \ \leq \ \bar{q}_1 \ \leq \ 1.917}$$

Footnote 5 (p. 332): each interval is **symmetric around that type's monopoly output** (1.5 for H, 2 for L), because each type's best deviation is its monopoly output.

Pooling deters nothing — the entrant enters anyway — so in pooling the "limit price" buys nothing; Tadelis calls pooling "somewhat artificial" (p. 332), and §16.3 makes that precise.

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5 · U — political incumbents (p. 332)

Epstein and Zemsky (1995): strong incumbents raise large campaign war chests as a **signal of strength** to deter challengers — the same entry-deterrence logic with money instead of output.

Links

- Prev: L08-06-spence-job-market · Next: L08-08-intuitive-criterion (§3: why only $q_1^{1L} = 2.62$ survives).
- Cournot best replies and the Bayesian Cournot game: Lecture 2, Lecture 7.
- Entry deterrence by commitment rather than information: Lecture 3; project practice `Listas/pratica-aula-01-02.md` item 18 (M&R FFT 5.2, p. 164).
- Code: every number of this note, asserted, in L08-11-python §7.
- Exercises (number and page only): Tadelis 16.1, 16.2, 16.3 (p. 335), 16.7 (p. 338); Muñoz-Garcia Ch. 9 Exercise 6 (p. 432); Ch. 11 Exercises 5 (p. 506) and 6 (p. 510).

16.3 · The intuitive criterion: read a deviation as coming from whoever could gain by it

> **Claim.** PBE lets off-path beliefs be anything, so signaling games have **continua** of equilibria. Cho and Kreps's **intuitive criterion** disciplines those beliefs: if an unused message could **never** help some type relative to its equilibrium payoff (it is *equilibrium-dominated* for that type), the Receiver should not attribute it to that type — and if, so attributed, the message makes another type **strictly better off for sure**, the equilibrium is broken. In the two-type games of this lecture it **removes every pooling PBE and every separating PBE but the least-cost one**: MBA $\to (DU, BM)$; Spence $\to e_H = 1.5$; limit pricing $\to q_1^{1L} = 2.62$.

Prev: L08-07-limit-pricing · Next: L08-09-formulas · Map: L08-00-map

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1 · U — the speech (p. 333)

In the MBA pooling equilibrium (UU, BB) (L08-05-mba-game §5.2) both types get 6. A high type deviates to D and says:

> *"I am an H type. To convince you I am going to deviate and choose D. If you believe me, and put me in the M job instead of a B job, I will get 8 instead of 6. The reason you should believe me is that if I were an L type who chose D and you were to promote me then I would get 5 instead of 6. Therefore you should believe me … because no L type in his right mind would do what I am about to do."*

The employer can make the deduction alone (p. 333): *if only one type can benefit from the deviation and the others can only lose, update toward that type.* This is **forward induction** — reading the past move as rational — as opposed to backward induction, which assumes future moves are rational (Gibbons p. 239).

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2 · C — the definition, and the MBA pooling equilibrium fails it (pp. 333–334)

Types Θ (finite), Sender's action a_1, Receiver's action a_2. For a subset of types $\hat{\Theta} \subseteq \Theta$ and an action a_1, let

$$BR_2(\hat{\Theta}, a_1) = \bigcup_{\mu:\ \mu(\hat{\Theta})=1} \arg\max_{a_2 \in A_2} \sum_{\theta \in \hat{\Theta}} v_2(a_1, a_2; \theta)\, \mu(\theta)$$

— the Receiver's best replies to a_1 under **some** belief concentrated on $\hat{\Theta}$. Write $v_1^*(\theta)$ for type θ's payoff in the PBE σ^*.

> **Definition 16.1.** A PBE σ^* **fails the intuitive criterion** if there exist an action a_1, a set of types $\Theta' \subset \Theta$ and a type $\theta' \notin \Theta'$ such that 1. $v_1^*(\theta) > \max_{a_2 \in BR_2(\Theta, a_1)} v_1(a_1, a_2; \theta)$ for all $\theta \in \Theta'$; 2. $v_1^*(\theta') < \min_{a_2 \in BR_2(\Theta \setminus \Theta', a_1)} v_1(a_1, a_2; \theta')$.

(The book prints v_2 inside the minimum in condition 2; it must be the Sender's payoff v_1, as the explanation on pp. 333–334 makes clear.)

In words. (1) Every type in Θ' does strictly worse with a_1 than in equilibrium **whatever the Receiver believes** — a_1 is *equilibrium-dominated* for Θ'. (2) Type θ' does strictly better with a_1 than in equilibrium **whatever the Receiver does**, as long as it believes the sender is not in Θ'.

The four-step test (two types):

Step	MBA pooling (UU, BB), off-path $a_1 = D$
1. Equilibrium payoffs	$v^*(H) = 6$, $v^*(L) = 6$
2. Best payoff from a_1 under any belief (M or B are both possible replies)	H: $\max(10-2,\, 6-2) = 8$; L: $\max(10-5,\, 6-5) = 5$
3. Who is equilibrium-dominated?	L: $5 < 6 \Rightarrow \Theta' = \{L\}$. H: $8 \geq 6$, kept
4. Belief on the kept types only ($\mu_D = 1$) → reply M; does a kept type gain for sure?	H gets $8 > 6 \Rightarrow$ **fails**

The separating PBE (DU, BM) **passes trivially**: both messages are used, so there is no off-path action to deviate to, and no type can gain "by convincing player 2 that he is truly that type" (p. 334). So the criterion selects (DU, BM).

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3 · C — limit pricing (pp. 334–335)

The entrant's best replies to any first-period output: at belief "high cost" it enters with $q_2^2 = 1$; at "low cost" it stays out ($q_2^2 = 0$); at intermediate beliefs it enters with smaller quantities, down to the level $\underline{q} < \frac{5}{6}$ at which entry just breaks even. So $BR_2(\Theta, a_1) = \{0\} \cup [\underline{q}, 1]$ for every a_1.

Pooling fails. Take $a_1 = 3$, $\Theta' = \{H\}$, $\theta' = L$.

- (1) H: pooling gives at least 3.25 (its (ICH) constraint), while producing 3 gives at most $(5-3-2)\cdot 3 + 2.25 = 0 + 2.25 = 2.25 < 3.25$ — even with no entry. **Equilibrium-dominated.**
- (2) L, believed to be L, faces no entry: $(5-3-1)\cdot 3 + 4 = 3 + 4 = 7$, while its **best** pooling payoff is $4 + (\frac{19}{12})^2 = 4 + 2.507 = 6.507$. $7 > 6.507$:

$$(5-3-1)3 + 4 = 7 \;>\; 4 + \left(\tfrac{19}{12}\right)^2 \qquad (16.3)$$

Every pooling PBE fails.

Only the best separating PBE survives (p. 335): of the separating equilibria $2.62 \leq q_1^{1L} \leq 3.32$, only $q_1^{1L} = 2.62$ passes. For any $q_1^{1L} > 2.62$, an output just above 2.62 is one H would never choose (it gives H less than 3.25 even with no entry — that is what 2.62 means) and that L, believed, strictly prefers (closer to its monopoly output 2). Tadelis leaves the formal check to Exercise 16.7 (p. 338).

> **Tadelis's summary (p. 335):** "Refinements such as the intuitive criterion help pin down equilibria, often resulting in the least-cost separating equilibrium."

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4 · Worked — Gibbons's examples

4.1 · Beer and Quiche (Gibbons pp. 237–239, Fig. 4.4.3)

Sender: wimpy (prob. 0.1) or surly (0.9). Message: Beer or Quiche for breakfast. Receiver: duel or not. Sender payoffs: preferred breakfast (wimpy: Quiche; surly: Beer) is worth 1, avoiding a duel 2. Receiver: duel a wimpy +1, duel a surly −1, not duel 0.

Pooling on Quiche, Receiver (not after Quiche, duel after Beer). After Quiche the belief is the prior: duel pays $0.1(1) + 0.9(-1) = -0.8 < 0$, so *not*. Equilibrium payoffs: wimpy $1 + 2 = 3$, surly $0 + 2 = 2$. After the unused Beer, duel needs $q = \Pr\{\text{wimpy} \mid \text{Beer}\}$ with $q(1) + (1-q)(-1) \geq 0$, i.e. $q \geq \frac{1}{2}$. Then Beer gives wimpy 0 and surly 1: no deviation. A PBE for any $q \geq \frac{1}{2}$ (p. 238).

It even passes Gibbons's **Signaling Requirement 5** (no *strictly dominated* message): Beer is not dominated for the wimpy type, since its best payoff from Beer (2) exceeds its worst from Quiche (1). But:

	Equilibrium payoff	Best payoff from Beer (any reply)	Equilibrium-dominated?
wimpy	3	2 (not dueled)	**yes**: $2 < 3$
surly	2	3 (not dueled)	no

Remove the wimpy type: $q = 0 \Rightarrow$ the Receiver does not duel after Beer $\Rightarrow$ surly gets $3 > 2$. **The pooling-on-Quiche PBE fails the intuitive criterion** (p. 239) — the surly type's "speech" is believable. (The same game, with the Receiver's payoffs shifted, is Tadelis Exercise 16.8, p. 338; this note works only the pooling-on-Quiche case that Gibbons works in his text.)

Equilibrium dominance vs dominance (Gibbons p. 239). A message is *dominated* for a type if another message's worst payoff beats its best payoff (Requirement 5); it is *equilibrium-dominated* if the type's **equilibrium payoff** beats its best payoff (Requirement 6 = the intuitive criterion). Dominated $\Rightarrow$ equilibrium-dominated, so Requirement 6 is stronger.

4.2 · Check: Gibbons's Fig. 4.2.2 pooling equilibrium (derived here)

L08-04-signaling-games §3 found the pooling PBE $[(L, L), (u, d), p = 0.5, q \leq \frac{2}{3}]$ with payoffs t_1: 1, t_2: 2. The unused message R: t_1 gets at most $\max(2, 0) = 2 \geq 1$ (kept); t_2 gets at most $\max(1, 1) = 1 < 2$ (**equilibrium-dominated**). Believing t_1 ($q = 1$), the Receiver plays u after R ($1 > 0$), and t_1 gets $2 > 1$. **This pooling PBE fails too**; the separating $[(R, L), (u, u)]$ uses both messages and passes.

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5 · C/G — Spence: only the least-cost separating equilibrium survives (Gibbons pp. 239–244; MWG pp. 468–471)

With M&R's numbers (L08-06-spence-job-market §3–4), both kinds of PBE are broken the same way: find an education level that **the low type would refuse even at wage θ_H** but that **the high type would take if paid θ_H**.

Separating at $e_H > 1.5$, e.g. $e_H = 2$: equilibrium payoffs L 20, H $50 - 20 = 30$. Any $e \in (1.5, 2)$: L's best is $50 - 20e < 50 - 30 = 20$ — equilibrium-dominated; H believed high gets $50 - 10e > 50 - 20 = 30$. Broken. At $e_H = 1.5$ there is no such gap: $(1.5, 1.5)$ is empty.

Pooling at e_p (wage $\bar{w} = 29$): $u_L^* = 29 - 20e_p$, $u_H^* = 29 - 10e_p$. A deviation e breaks it if

$$\underbrace{50-20e<u_L^*}_{\text{L excluded}} \quad \text{and} \quad \underbrace{50-10e>u_H^*}_{\text{H gains}} \iff e\in\left(\frac{50-u_L^*}{20},\ \frac{50-u_H^*}{10}\right)=\left(\tfrac{21}{20}+e_p,\ \tfrac{21}{10}+e_p\right),$$

which is never empty because $c_H < c_L$ and $\theta_H > \bar{w}$. At $e_p = 0.3$: (1.35, 2.4).

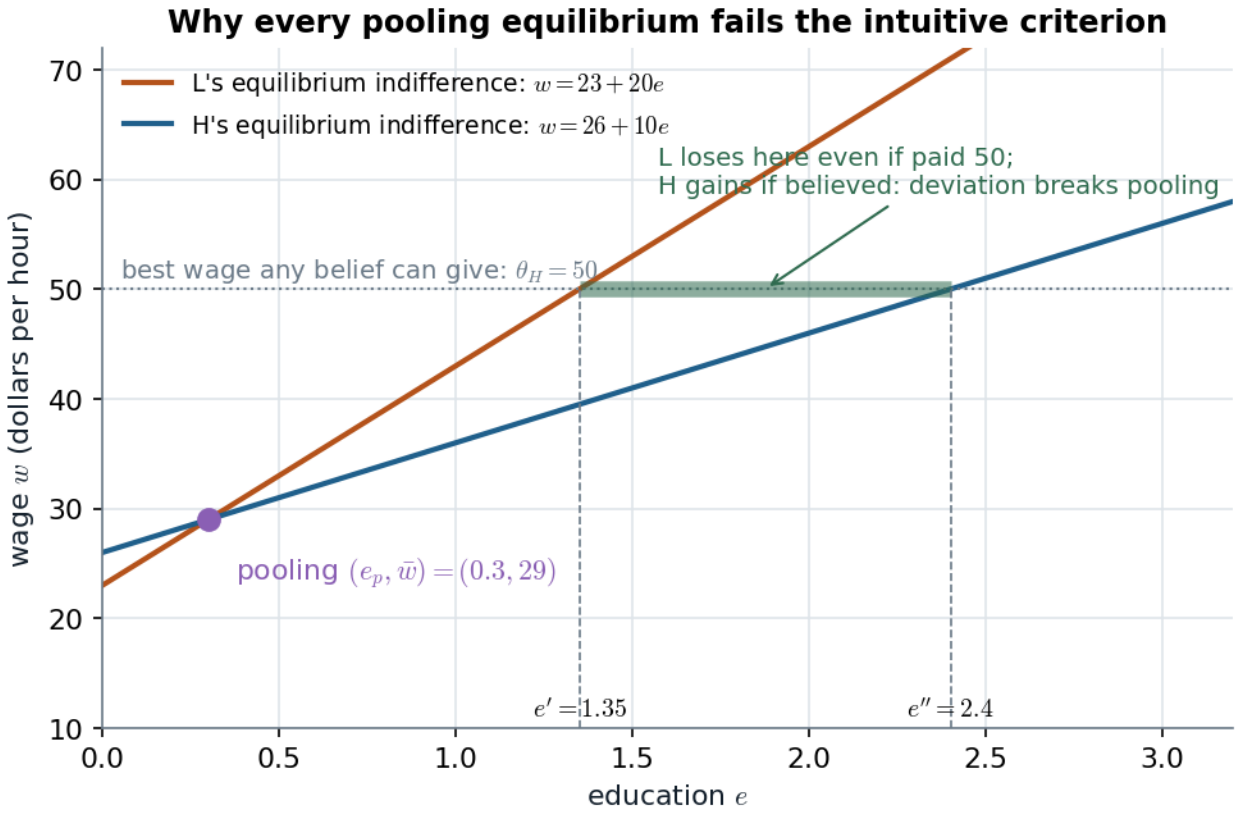


Figure — why pooling at $e_p = 0.3$ fails: between $e' = 1.35$ and $e'' = 2.4$ the low type is worse off than in equilibrium even at the best possible wage, while the high type gains if recognized. Source: Gibbons pp. 239–244 (Figs. 4.4.4–4.4.5); MWG Fig. 13.AA.2 (p. 470); M&R numbers.

The gap opens because the two equilibrium indifference lines through the pooling point have different slopes — single crossing again. Any pooling point has such a gap, so **no pooling PBE survives**, and the only survivor is the separating equilibrium in which the low type is exactly indifferent to mimicking: $e_H = (\theta_H - \theta_L)/c_L = 1.5$ (Gibbons Fig. 4.4.4: "just enough education to make the low-ability worker indifferent about mimicking"). The Education Signal companion (`companions/companion-spence-signaling.html`) reports this survivor as its headline number.

> **Interactive companion.** The MBA Game (`companions/companion-mba-game.html`) — the last column of its table applies this test to every PBE; drag c_L below $w_M - w_B$ and the pooling PBE *survives*, because then D is no longer equilibrium-dominated for the low type.

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6 · G — MWG Appendix 13.AA (pp. 467–471) and forward induction

MWG builds the refinement in four steps (pp. 468–471):

Eq.	Test on action a for type θ	What it removes in the education model
(13.AA.1)	strictly dominated: some a' has $\min_{s'} u_1(a', s', \theta) > \max_s u_1(a, s, \theta)$ over **all** responses	—
(13.AA.2)	the same, but only over **equilibrium** responses $S^*(\Theta, a)$	every separating PBE but the best; pooling PBE that H likes less than $(\theta_H, \tilde{e})$
(13.AA.3)	**equilibrium dominated**: $u_1^*(\theta) > \max_{s \in S^*(\Theta,a)} u_1(a, s, \theta)$	also **every** pooling PBE (Fig. 13.AA.2, p. 470)
(13.AA.4)	intuitive criterion: some type has $\min_s u_1(a, s, \theta) > u_1^*(\theta)$ once equilibrium-dominated types are excluded	leaves "the best separating equilibrium"

With **two types** the intuitive criterion and the reasonable-beliefs refinement based on (13.AA.3) coincide (p. 471); MWG's Example 13.AA.1 (three types) shows a PBE that passes the intuitive criterion but fails the reasonable-beliefs test — named only. The general idea of disciplining off-path beliefs by the deviator's rationality is MWG's **forward induction** (§9.D, p. 292). Cho and Kreps also show every signaling game in this class has a PBE satisfying the criterion (Gibbons p. 239).

Links

- Prev: L08-07-limit-pricing · Next: L08-09-formulas.
- The equilibria being refined: L08-05-mba-game, L08-06-spence-job-market, L08-07-limit-pricing; the first refinement: L08-03-sequential-equilibrium.
- Rules: rules/game-theory item 3 (state the selection argument) and item 18 (do not pick the refinement after seeing the answer).
- Code: `intuitive_fails`, tested on the MBA game, Gibbons Fig. 4.2.2 and Beer–Quiche, in L08-11-python §5.
- Exercises (number and page only): Tadelis 16.7, 16.8 (p. 338); Muñoz-Garcia Ch. 9 Exercise 9 (p. 447), Ch. 11 Exercise 4 (p. 500); MWG 13.AA.1 (p. 476).

Lecture 8 · Formulas — every definition, every result, every number

Prev: L08-08-intuitive-criterion · Next: L08-10-pitfalls · Map: L08-00-map

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0 · The exam specification, mapped

Estimated unit: no slides yet, so the specification is the study guide's Lecture 8 block (`Map/study-guide.md`) and the chapters it assigns, item by item.

#	Item (study guide / chapter content)	Note
1	Why subgame perfection fails under incomplete information (Tadelis §15.1)	L08-01-subgame-perfection-fails
2	Beliefs, Bayes' rule on the path, anything off the path; sequential rationality; PBE (Tadelis §15.2; Gibbons §4.1)	L08-02-beliefs-and-pbe
3	Prop. 15.1 — a BNE that reaches every information set is a PBE	L08-02-beliefs-and-pbe §6
4	Sequential equilibrium and consistency (Tadelis §15.3)	L08-03-sequential-equilibrium
5	Signaling games: pooling, separating, semi-separating; the method (Tadelis Ch. 16 opening; Gibbons §4.2.A)	L08-04-signaling-games
6	The MBA game: matrix, BNE, PBE, supporting beliefs (Tadelis §16.1)	L08-05-mba-game
7	Education signaling à la Spence (Gibbons §4.2.B; M&R p. 154)	L08-06-spence-job-market
8	Limit pricing and entry deterrence (Tadelis §16.2)	L08-07-limit-pricing
9	Refinements: the intuitive criterion (Tadelis §16.3; Gibbons §4.4)	L08-08-intuitive-criterion
10	Formal side (MWG §9.C, §9.D, §13.C, 13.AA) — **G**, named	notes 02 §7, 03 §4, 06 §7, 08 §6
11	Applications: signaling vs screening (M&R pp. 154–157); adverse selection link	L08-06-spence-job-market, L08-10-pitfalls Part I §8

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1 · Definitions and results

1.1 · Tadelis, numbered

Item	Statement	Note
Def. 15.1 (p. 307)	An information set is **on** the path of a BNE σ^* if reached with positive probability, **off** otherwise	02 §1
Def. 15.2 (p. 308)	A **system of beliefs** μ: a distribution over nodes of every information set, $\sum_{x \in h} \mu(x) = 1$	02 §2
Req. 15.1 (p. 308)	every player has a belief at each of his information sets	02 §2
Req. 15.2 (p. 309)	on the path, beliefs follow **Bayes' rule**	02 §3
Req. 15.3 (p. 310)	off the path, **any** belief	02 §4
Req. 15.4 (p. 310)	**sequential rationality**: $E[v_i(\sigma_i^*, \sigma_{-i}^*, \theta_i) \mid h, \mu] \geq E[v_i(s_i, \sigma_{-i}^*, \theta_i) \mid h, \mu]$ at every h	02 §4
Def. 15.3 (p. 310)	**PBE** = BNE σ^* + beliefs μ satisfying Reqs. 15.1–15.4	02 §4
Prop. 15.1 (p. 311)	a BNE reaching every information set, with Bayes beliefs, is a PBE	02 §6
Def. 15.4 (p. 313)	**consistency**: $(\sigma^*, \mu^*) = \lim(\sigma^k, \mu^k)$, σ^k totally mixed, μ^k by Bayes	03 §2
Def. 15.5 (p. 313)	**sequential equilibrium** = consistent PBE	03 §2
(16.1)–(16.2) (p. 321)	$\mu_U = \frac{p\sigma_H}{p\sigma_H + (1-p)\sigma_L}$, $\mu_D = \frac{p(1-\sigma_H)}{p(1-\sigma_H) + (1-p)(1-\sigma_L)}$	05 §2
Claims 16.1–16.2 (p. 326)	separating limit pricing: entry iff believed H; $q_1^{1H} = 1.5$	07 §3
Def. 16.1 (p. 333)	a PBE **fails the intuitive criterion** if some a_1 is equilibrium-dominated for Θ' and, believed not in Θ', gives some $\theta' \notin \Theta'$ more for sure	08 §2
(16.3) (p. 334)	$(5-3-1)3+4 = 7 > 4+(19/12)^2$: L's deviation breaks limit-pricing pooling	08 §3

1.2 · Gibbons

Item	Statement	Note
Requirements 1–4 (pp. 177–180)	belief at each information set; sequential rationality; Bayes on the path; Bayes **where possible** off it	02 §7
Signaling Reqs. 1, 2R, 2S, 3 (pp. 187–188)	Receiver belief; Receiver best reply; Sender best reply; Bayes on used messages	04 §2
Single crossing (p. 192)	$c_e(L,e) > c_e(H,e)$	06 §1
(4.2.1) (p. 193)	$w(e) = \mu(H \mid e)y(H,e) + [1-\mu(H \mid e)]y(L,e)$	06 §1
(4.2.2) (p. 196)	pooled wage $w_p = q\,y(H,e_p) + (1-q)\,y(L,e_p)$	06 §4
(4.2.8)–(4.2.10) (pp. 202–203)	hybrid: $r = \dfrac{q}{q+(1-q)\pi}$; L indifferent; $w_h = r\,y(H,e_h) + (1-r)\,y(L,e_h)$ $\Rightarrow \pi = \dfrac{q(1-r)}{r(1-q)}$	06 §6
(4.2.11)–(4.2.13) (p. 206)	Myers–Majluf: accept iff $s[qL+(1-q)H+R] \geq I(1+r)$; go ahead iff $s \leq R/(\pi+R)$; pooling iff $\dfrac{I(1+r)}{pL+(1-p)H+R} \leq \dfrac{R}{H+R}$	04 §5
Signaling Reqs. 5–6 (pp. 237, 239)	no belief on a type for which the message is **dominated** (5) / **equilibrium-dominated** (6 = intuitive criterion)	08 §4

1.3 · Unnumbered but examinable

Result	Statement	Note
MBA employer's threshold	$M \iff 10\mu \geq 3+2\mu \iff \mu \geq \frac{3}{8}$	05 §3
MBA separation	$c_H \leq w_M - w_B \leq c_L$ $(2 \leq 4 \leq 5)$	05 §5.1
Spence separating set (linear cost)	$e_H \in \left[\dfrac{\theta_H-\theta_L}{c_L},\ \dfrac{\theta_H-\theta_L}{c_H}\right]$, $e_L = 0$	06 §3
Least-cost separating	$e^* = \dfrac{\theta_H-\theta_L}{c_L}$ — L exactly indifferent to mimicking	06 §3, 08 §5
Spence pooling set (belief "low" off path)	$e_p \in \left[0,\ \dfrac{p(\theta_H-\theta_L)}{\max(c_H,c_L)}\right]$	06 §4
H prefers signaling to no signaling	iff $p \leq p^* = 1 - \dfrac{c_H}{c_L}$	06 §5
Expected signaling waste	$p\,c_H\,e^*$ per worker	06 §5

Result	Statement	Note
Intuitive criterion in Spence	pooling at e_p broken by $e \in \left(\frac{\theta_H - u_L^*}{c_L}, \frac{\theta_H - u_H^*}{c_H}\right)$	08 §5

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2 · Every number

2.1 · Entry game (Tadelis pp. 304–311)

Quantity	Value
Payoffs: O; C fought / accommodated; W fought / accommodated	$(0,2)$; $(-1,-1)/(1,1)$; $(-2,0)/(-1,1)$
(OE, A), general p	$(p-1,\ 1+p)$ (book prints $1-p$ for the second)
(EE, F), general p	$(p-2,\ -p)$
Pure BNE at $p = \frac{1}{2}$	(OO, F), (EO, A)
After entry	$E[v_2(A)] = 1 > E[v_2(F)] = -\mu$ for all μ
Unique PBE	(EO, A), $\mu(x_3) = 1$
Mixed BNE	OO with $\Pr\{F\} \geq \frac{1}{2}$

2.2 · Gibbons Fig. 4.1.1 and Tadelis Fig. 15.4

Quantity	Value
Fig. 4.1.1: $E[u_2(L')]$ vs $E[u_2(R')]$	$2-p > 1-p$ — (R, R') dies
Fig. 15.4: R optimal iff	$3\mu \geq 2$, $\mu \geq \frac{2}{3}$
Fig. 15.4: consistent belief	$\mu_2(x_1) = \frac{1}{2} \Rightarrow$ unique SE (D, L)

2.3 · Gibbons Fig. 4.2.2 (pp. 188–190)

Quantity	Value
After L	u at every p $(4-p > 1-p)$
After R	d iff $q \leq \frac{2}{3}$
PBE	pooling $[(L,L),(u,d),p=0.5,q\leq\frac{2}{3}]$; separating $[(R,L),(u,u),p=0,q=1]$
Intuitive criterion (derived, note 08 §4.2)	the pooling PBE fails

2.4 · MBA game (Tadelis pp. 319–323, 333–334)

Quantity	Value
Costs, wages, prior	$c_H = 2$, $c_L = 5$, $w_M = 10$, $w_B = 6$, $p = \frac{1}{4}$
Employer profits H: M, B; L: M, B	$10, 5$; $0, 3$
(UD, MB)	$(3.25,\ 4.75)$
(DU, BB)	$(5.5,\ 3.5)$ (book prints 4.5)

Quantity	Value
Pure BNE	(UU, BB), (DU, BM)
Separating PBE	(DU, BM), $\mu_U = 0$, $\mu_D = 1$; payoffs H 8, L 6
Pooling PBE	(UU, BB), $\mu_U = \frac{1}{4}$, $\mu_D \in [0, \frac{3}{8}]$; payoffs 6, 6
Intuitive criterion	pooling fails (L's best from D is $5 < 6$; H believed gets $8 > 6$)

2.5 · Spence with M&R's numbers (pp. 155–156)

Quantity	Value
θ_H, θ_L, share high, c_H, c_L	\$50, \$20, 30%, \$10, \$20
Pooled wage	$0.3(50) + 0.7(20) = \$29$
Separating e_H	$[1.5,\ 3]$; M&R's check at $E_H = 2$: L nets \$10 < \$20, H nets \$30 > \$20
Least-cost e^*, H's net pay	1.5, \$35
Pooling e_p	$[0,\ 0.45]$
p^*	0.5
Waste per worker	$0.3 \times 10 \times 1.5 = \$4.50$
Hybrid at $e_h = 1$	$w_h = 40$, $r = \frac{2}{3}$, $\pi = 0.214$; hybrids for $e_h \in [0.45, 1.5]$
Intuitive criterion at $e_p = 0.3$	breaking deviations $e \in (1.35,\ 2.4)$

2.6 · Limit pricing (Tadelis pp. 323–335)

Quantity	Value
Demand, costs, entry cost, prior	$P = 5 - q$; $c_1 \in \{1, 2\}$, $c_2 = 2$; $F = \frac{1}{2}$; $\frac{1}{2}$
Monopoly L, H	$q = 2$, profit 4; $q = 1.5$, profit 2.25
Cournot, known H	$q_1 = q_2 = 1$, $P = 3$, $v_1 = 1$, $v_2 = \frac{1}{2}$
Cournot, known L	$q_1 = \frac{5}{3}$, $q_2 = \frac{2}{3}$, $P = \frac{8}{3}$, $v_1 = \frac{25}{9}$, $v_2 = -\frac{1}{18}$
H's separating payoff; mimicking at 2	3.25; 4.25
L's best deviation	$4 + 2.25 = 6.25$
Separating q_1^{1L}	$[\frac{3+\sqrt{5}}{2},\ 2 + \sqrt{1.75}] = [2.62,\ 3.32]$
Bayesian Cournot after pooling	$q_1^{2L} = \frac{19}{12}$, $q_1^{2H} = \frac{13}{12}$, $q_2^2 = \frac{5}{6}$; entrant $\frac{7}{36}$ (book prints $\frac{7}{18}$)
(ICH), (ICL)	$[1.083,\ 1.917]$, $[1.493,\ 2.507]$
Pooling $\bar{q}_1$	$[1.493,\ 1.917]$
Intuitive criterion	pooling broken by $q = 3$; only $q_1^{1L} = 2.62$ survives

2.7 · Beer and Quiche (Gibbons pp. 237–239)

Quantity	Value
Prior wimpy, surly	0.1, 0.9
Pooling on Quiche	not after Quiche; duel after Beer iff $q \geq \frac{1}{2}$; payoffs wimpy 3, surly 2

Quantity	Value
Intuitive criterion	Beer equilibrium-dominated for wimpy ($2 < 3$), not for surly $\Rightarrow q = 0 \Rightarrow$ surly gets $3 > 2$: fails

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Links

- Map: L08-00-map · Pitfalls and self-test: L08-10-pitfalls · Code: L08-11-python.
- Rules: rules/07 · rules/game-theory.

Lecture 8 · Pitfalls and an original True/False self-test

Prev: L08-09-formulas · Next: L08-11-python · Map: L08-00-map

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Part I — the confusions this material produces

1 · "A PBE is a strategy profile"

It is a profile *and* a belief system. Stating (UU, BB) alone is not a PBE of the MBA game; (UU, BB) with $\mu_U = \frac{1}{4}$ and $\mu_D = 0.9$ is **not** one (at $\mu_D = 0.9 > \frac{3}{8}$ the employer would promote after an MBA). The beliefs are part of the answer; "$\mu_D \leq \frac{3}{8}$" is what makes it an equilibrium. → L08-05-mba-game §5.2.

2 · "Off the path, beliefs must be the prior"

No — off the path Bayes' rule is $\frac{0}{0}$ and Requirement 15.3 allows any belief. On the path in a *pooling* equilibrium the belief equals the prior; at an *unused* message it is free. Confusing the two makes one wrongly "derive" $\mu_D = p = \frac{1}{4}$ in the MBA pooling equilibrium (it happens to support B, but $\mu_D = 0.3$ would too, and $\mu_D = 0.5$ would not). → L08-02-beliefs-and-pbe §3–4.

3 · "Subgame perfection already handles credibility"

Not with private types. In the entry game with types, the incumbent's nodes after entry form one information set; the only proper subgame is the whole game, so the non-credible (OO, F) is subgame perfect. Only a belief at the information set kills it. → L08-01-subgame-perfection-fails §3.

4 · "Separating means the high type does what it would do under full information"

In the envy case it over-invests. With unproductive education, full information gives $e^*(H) = 0$; separation needs $e_H \geq 1.5$ (M&R numbers). The low type is the one who keeps his full-information choice ($e_L = 0$, MWG Lemma 13.C.2). → L08-06-spence-job-market §2–3.

5 · "Education raises wages, so it must raise productivity"

False in the pure Spence model: output is θ whatever the education, yet wages rise with education in a separating equilibrium because education **reveals** θ. The rules style guide uses exactly this trap (`rules/00_estilo_avaliacao.md` §2). → L08-06-spence-job-market §1.

6 · "Signaling makes the able better off"

Not necessarily. The low type is always worse off ($20 < 29$); the high type is better off only if $p \leq p^* = 1 - c_H/c_L$. With M&R's costs and 70% high types, the pooled wage $20 + 30(0.7) = 41$ beats the 35 a high type nets after signaling. → L08-06-spence-job-market §5.

7 · "Any cost difference makes a signal work"

What matters is the marginal cost, and its ranking. If $c_H \geq c_L$ the separating interval $[\Delta/c_L, \Delta/c_H]$ is empty: any education the high type would buy, the low type buys too (M&R p. 156). In the MBA game, separation needs $c_H \leq w_M - w_B \leq c_L$: with $c_L = 3 < 4$, the low type mimics and only pooling survives. → L08-05-mba-game §5.1.

8 · Signaling vs screening

Who moves first. Signaling: the **informed** party acts (education, output, advertising) and the uninformed infers. Screening: the **uninformed** party offers a menu and the informed self-select (M&R pp. 154, 156–157; e.g. deductible vs no-deductible insurance). The self-selection constraints look alike; the timing and the solution concept differ (screening is a design problem — out of this course's scope as mechanism design).

9 · "The intuitive criterion picks the equilibrium the Sender likes best"

It picks the one that survives a credible-deviation test. In the two-type games here that is the least-cost separating equilibrium, but that need not be the high type's favourite: at $p > p^*$ the high type prefers the pooling outcome the criterion eliminates (note 06 §5). → L08-08-intuitive-criterion.

10 · "Pooling on the off-path message tells you nothing, so ignore it"

The reply to the unused message is the equilibrium's enforcement device. In Gibbons's Fig. 4.2.2, pooling on L requires the reply d after R; with u after R, type t_1 deviates ($2 > 1$). Writing the receiver strategy for used messages only is an incomplete strategy. → L08-04-signaling-games §3.

Part II — True/False self-test (original items)

Scoring as in `rules/00_estilo_avaliacao.md`: score = $\max\{0,\ \text{correct} - \text{wrong}/2\} \times$ item value; blanks are not wrong. **[E]** easy · **[M]** medium · **[P]** hard.

Block A — beliefs and PBE

1. In a perfect Bayesian equilibrium, beliefs at every information set must follow Bayes' rule. **[E] F.** Only on the equilibrium path (Requirement 15.2). Off the path Bayes' rule gives 0/0 and Requirement 15.3 allows any belief — which is precisely what lets pooling equilibria exist.

2. If a Bayesian Nash equilibrium reaches every information set with positive probability, it is a PBE once beliefs are computed by Bayes' rule. **[E] T.** Proposition 15.1: any failure of sequential rationality at a reached information set would be a profitable ex-ante deviation, contradicting BNE.

3. In the entry game with a weak and a competitive entrant, the BNE in which both types stay out is subgame perfect but not a PBE. **[M] T.** The only proper subgame is the whole game, so the BNE (OO, F) is subgame perfect; but after entry A yields 1 and F yields $-\mu \leq 0$, so no belief makes F a best reply.

4. A strategy that is never played on the equilibrium path cannot affect whether a profile is a PBE. **[E] F.** Off-path actions must be sequentially rational given some belief, and they deter deviations: the employer's B after an unused MBA is what sustains the MBA pooling equilibrium.

5. Every sequential equilibrium is a PBE, but some PBE are not sequential equilibria. **[M] T.** Sequential equilibrium adds consistency (Def. 15.5). In Tadelis's Fig. 15.4, (U, R) is a PBE with $\mu > \frac{2}{3}$, but trembles force $\mu = \frac{1}{2}$, under which R is not optimal.

6. Consistency requires off-path beliefs to equal the prior distribution of Nature's moves. **[P] F.** It requires them to be limits of Bayes' rule under totally mixed strategies; they equal the prior in Fig. 15.4 only because player 1 cannot condition on Nature. In general the trembles of players who do observe Nature can shift the odds.

Block B — signaling games

7. In a pooling equilibrium of a signaling game, the receiver's belief after the equilibrium message equals the prior. **[E] T.** All types send it, so Bayes' rule gives $\mu(t \mid m) = p(t)$: the message carries no information.

8. In a separating equilibrium with two types and two messages, the receiver's beliefs are unrestricted at some information set. **[M] F.** Both messages are used, so both beliefs are pinned (0 and 1). Free beliefs arise in separating equilibria only when there are more messages than types.

9. In Tadelis's MBA game at the book's numbers, the employer promotes a worker after an MBA whenever its belief that he is high-skill is at least 3/8. **[E] T.** $10\mu \geq 5\mu + 3(1-\mu) \iff 8\mu \geq 3$.

10. The MBA game's separating equilibrium exists for every prior $p \in (0,1)$, holding costs and wages at the book's values. **[M] T.** It uses beliefs 0 and 1 and the constraints $c_H \leq w_M - w_B \leq c_L$ $(2 \leq 4 \leq 5)$, none of which involves p.

11. In the MBA game, the pooling equilibrium in which no one studies would still exist if the employer believed that an MBA holder is high-skill with probability 0.5. **[M] F.** At $\mu_D = 0.5 > \frac{3}{8}$ the employer promotes after an MBA; the high type then gets $10 - 2 = 8 > 6$ from studying and deviates.

12. For a costly action to work as a signal, it must be cheaper for the type that wants to be identified. **[E] T.** Otherwise the other type mimics (M&R p. 156: if $C_H \leq C_L$ fails the self-selection inequalities cannot both hold); formally single crossing, $c_e(L,e) > c_e(H,e)$.

Block C — Spence

13. In Spence's model with unproductive education, a separating equilibrium raises wages with education and therefore raises output with education. **[E] F.** Output is θ regardless of e; wages differ because education reveals θ.

14. With $\theta_H = 50$, $\theta_L = 20$, $c_H = 10$ and $c_L = 20$, any e_H between 1.5 and 3 supports a separating equilibrium. **[E] T.** L will not mimic iff $50 - 20e_H \leq 20$, i.e. $e_H \geq 1.5$; H prefers signaling iff $50 - 10e_H \geq 20$, i.e. $e_H \leq 3$.

15. In those separating equilibria the low-ability worker chooses a strictly positive amount of education. **[E] F.** He is paid θ_L whatever he does, so any $e > 0$ is a pure loss: $e_L = 0$ (MWG Lemma 13.C.2).

16. A higher share of high-ability workers raises the least-cost separating education level. **[M] F.** $e^* = (\theta_H - \theta_L)/c_L$ does not involve the prior: the separating set is independent of p (MWG p. 456). The prior moves the pooled wage and p^*-comparisons only.

17. With M&R's numbers and 70% high-ability workers, high-ability workers are worse off in the least-cost separating equilibrium than if signaling were impossible. **[M] T.** Signaling nets them $50 - 10(1.5) = 35$; no signaling pays everyone $0.7(50) + 0.3(20) = 41 > 35$ $(p = 0.7 > p^* = 0.5)$.

18. In the separating equilibria of the Spence model, the expenditure on education is socially wasted when education is unproductive. **[E] T.** It changes no output, only who is paid what; M&R call it "a pure social waste" (p. 156).

19. In a hybrid equilibrium in which the low type randomizes, the belief after the high type's education level lies strictly between the prior and 1. **[M] T.** $\mu = \dfrac{q}{q + (1-q)\pi}$ with $\pi \in (0,1)$ lies in $(q,1)$ (Gibbons 4.2.8).

Block D — limit pricing and refinements

20. In Tadelis's limit-pricing game, the low-cost incumbent can separate by producing its monopoly output of 2 in the first period. **[M] F.** The high-cost type would mimic: $2 + 2.25 = 4.25 > 3.25$. Separation needs $q_1^{1L} \geq 2.62$.

21. In the separating equilibria of the limit-pricing game, the low-cost incumbent's first-period price is below its marginal cost. **[P] F.** At $q \in [2.62, 3.32]$ the price is $5 - q \in [1.68, 2.38] > 1$. Below-cost pricing needs a growing market or more deterrence periods (pp. 329–330).

22. In the pooling equilibria of the limit-pricing game, the entrant stays out. **[M] F.** With belief $\frac{1}{2}$ the Bayesian Cournot game gives the entrant $\frac{7}{36} > 0$, so it enters; pooling buys the incumbent no deterrence.

23. The intuitive criterion eliminates the MBA game's pooling equilibrium because the low type could never gain by getting an MBA, whatever the employer then believes. **[E] T.** L's best payoff from D is $10 - 5 = 5 < 6$; excluding L, the employer promotes after D and the high type gets $8 > 6$.

24. A message that is equilibrium-dominated for a type is necessarily dominated for that type. **[P] F.** The converse holds (dominated ⇒ equilibrium-dominated). In Beer–Quiche, Beer is equilibrium-dominated for the wimpy type ($2 < 3$) but not dominated (its best from Beer, 2, exceeds its worst from Quiche, 1) — Gibbons p. 239.

25. In the two-type Spence model with single crossing, every pooling equilibrium fails the intuitive criterion. **[M] T.** Between $(\theta_H - u_L^*)/c_L$ and $(\theta_H - u_H^*)/c_H$ there is always an education level the low type would refuse even at θ_H and the high type would take if believed (Gibbons pp. 239–244; MWG p. 471).

26. The intuitive criterion selects, among the separating equilibria of the limit-pricing game, the one with the largest first-period output. **[M] F.** It selects the **least-cost** one, $q_1^{1L} = 2.62$ — the smallest output that still deters mimicry (p. 335).

Links

- Map: L08-00-map · Formulas: L08-09-formulas · Code: L08-11-python.
- Style: `rules/00_estilo_avaliacao.md` (True/False rule) · Rules: rules/07.

Lecture 8 · Computation in Python

> Every routine the lecture needs — Bayes' rule at an information set, the Bayesian normal form, an exhaustive pure-PBE solver for two-type signaling games, the intuitive-criterion test, the Spence intervals, and the limit-pricing bounds — each with `assert` self-checks against the books' own numbers. Run the blocks in order (they share names); the last line prints `L08 reference implementation: all checks pass.` Python + numpy only.
>
> Prev: L08-10-pitfalls · Map: L08-00-map · Formulas: L08-09-formulas

Reuse. Nothing in `Listas/*-check.py` or `Aula/**/check_*.py` covers Bayesian updating or signaling (those files serve Lectures 1–2), so the routines are new. The companions' independent checks live in `companions/check_companions.py` (run: `python Notas/L08/companions/check_companions.py`).

Two things the code found in the book (both noted where they occur): the matrix cell (DU, BB) on p. 322 should read 5.5, not 4.5; the entrant's pooled-entry profit on p. 331 is 7/36, not 7/18. Neither changes a conclusion.

---

1 · Bayes' rule at an information set → L08-02-beliefs-and-pbe §3

Requirement 15.2 as a function. `None` is the off-path case of Requirement 15.3: no type sends the message, so Bayes' rule is silent.

```
from fractions import Fraction as Fr
from itertools import product


def posterior(prior, send):
    """Bayes' rule at the information set that follows one message.

    prior: {type: Pr(type)}; send: {type: Pr(type sends this message)}.
    Returns {type: posterior} or None when no type sends the message (off the path:
    Requirement 15.3, any belief is allowed).
    """
    den = sum(prior[t] * send[t] for t in prior)
    return None if den == 0 else {t: prior[t] * send[t] / den for t in prior}


p = Fr(1, 3)                                          # any prior works
assert posterior({"C": p, "W": 1 - p}, {"C": 1, "W": 0})["C"] == 1          # EO:
 ↪  entry => competitive
assert posterior({"C": p, "W": 1 - p}, {"C": 0, "W": 0}) is None            # OO:
 ↪  entry is off-path
sC, sW = Fr(1, 2), Fr(1, 4)                                                 # a
 ↪  behavioural mix
assert posterior({"C": p, "W": 1 - p}, {"C": sC, "W": sW})["C"] == p * sC / (p * sC
 ↪  + (1 - p) * sW)
```

---

2 · The entry game: Bayesian normal form, BNE, sequential rationality → L08-01-subgame-perfection-fails

The matrix printed on p. 306, the two pure BNE, and the one-line reason (OO, F) is not a PBE: A beats F at every belief.

```
# Tadelis Fig. 15.2 (p. 305): payoffs (entrant, incumbent) after (type, action[, response])
ENTRY = {("C", "O"): (0, 2), ("W", "O"): (0, 2),
         ("C", "E", "F"): (-1, -1), ("C", "E", "A"): (1, 1),
         ("W", "E", "F"): (-2, 0), ("W", "E", "A"): (-1, 1)}


def entry_payoffs(s1, s2, p):
    """Expected payoffs of the Bayesian normal form; s1 = (action of C, action of W), s2 in {F, A}."""
    v = [Fr(0), Fr(0)]
    for t, a, pr in [("C", s1[0], p), ("W", s1[1], 1 - p)]:
        out = ENTRY[(t, a)] if a == "O" else ENTRY[(t, a, s2)]
        v = [v[0] + pr * out[0], v[1] + pr * out[1]]
    return tuple(v)


half = Fr(1, 2)
S1 = [("O", "O"), ("O", "E"), ("E", "O"), ("E", "E")]
# the matrix printed on p. 306 at p = 1/2
assert entry_payoffs(("O", "E"), "F", half) == (-1, 1) and entry_payoffs(("O", "E"), "A", half) == (-half, Fr(3, 2))
assert entry_payoffs(("E", "O"), "F", half) == (-half, half) and entry_payoffs(("E", "O"), "A", half) == (half, Fr(3, 2))
assert entry_payoffs(("E", "E"), "F", half) == (Fr(-3, 2), -half) and entry_payoffs(("E", "E"), "A", half) == (0, 1)


def pure_nash(rows, cols, pay):
    """All pure Nash equilibria of a two-player matrix game given pay(r, c) -> (v1, v2)."""
    return [(r, c) for r in rows for c in cols
            if all(pay(r, c)[0] >= pay(r2, c)[0] for r2 in rows)
            and all(pay(r, c)[1] >= pay(r, c2)[1] for c2 in cols)]


bne = pure_nash(S1, ["F", "A"], lambda r, c: entry_payoffs(r, c, half))
assert sorted(bne) == sorted([(("O", "O"), "F"), (("E", "O"), "A")])        # p. 306
# sequential rationality after entry: A beats F at every belief mu = Pr(C | E)  (p. 310)
assert all(mu * 1 + (1 - mu) * 1 > mu * -1 + (1 - mu) * 0 for mu in [Fr(k, 100) for k in range(101)])
```

---

3 · An exhaustive pure-PBE solver for two-type signaling games → L08-04-signaling-games §3

For every Sender strategy and Receiver strategy: Bayes on used messages, an interval of supporting beliefs on unused ones, then the Sender's incentive constraints. Tested on Gibbons's Fig. 4.2.2, where it must find exactly the two PBE of pp. 188–190.

```
def br_interval(u2, m, a, acts, t1, t2):
    """Beliefs mu = Pr(t1 | m) in [0, 1] at which action a is a (weak) best response
    ↪ to message m.

    u2(t, m, a) is the receiver's payoff. Each comparison with another action is
↪   linear in mu,
    so the set is an interval; returns (lo, hi) or None if empty.
    """
    lo, hi = Fr(0), Fr(1)
    for b in acts:
        if b == a:
            continue
        # mu*d1 + (1-mu)*d2 >= 0, with d = u2(.,m,a) - u2(.,m,b)
        d1 = Fr(u2(t1, m, a) - u2(t1, m, b)); d2 = Fr(u2(t2, m, a) - u2(t2, m, b))
        slope, const = d1 - d2, d2
        if slope == 0:
            if const < 0:
                return None
        elif slope > 0:
            lo = max(lo, -const / slope)
        else:
            hi = min(hi, -const / slope)
    return (lo, hi) if lo <= hi else None


def pure_pbe(types, prior, msgs, acts, u1, u2):
    """Every pure-strategy PBE of a two-type signaling game (Tadelis Def. 15.3;
    ↪ Gibbons p. 188).

    types: (t1, t2); prior: Pr(t1); u1(t, m, a), u2(t, m, a): sender and receiver
↪   payoffs.
    Returns a list of (sender {t: m}, receiver {m: a}, beliefs {m: Pr(t1|m) or (lo,
↪   hi)}).
    """
    t1, t2 = types
    out = []
    for sm in product(msgs, repeat=2):
        s = dict(zip(types, sm))
        for ra in product(acts, repeat=len(msgs)):
            r = dict(zip(msgs, ra))
            beliefs, ok = {}, True
            for m in msgs:
                post = posterior({t1: prior, t2: 1 - prior}, {t: int(s[t] == m) for t
↪   in types})
```

```
                iv = br_interval(u2, m, r[m], acts, t1, t2)
                if post is None:                                    # off path: any
                 ↪ supporting belief
                    ok = iv is not None
                    beliefs[m] = iv
                else:                                               # on path: Bayes,
                 ↪ then optimality
                    mu = post[t1]
                    ok = iv is not None and iv[0] <= mu <= iv[1]
                    beliefs[m] = mu
                if not ok:
                    break
            if ok and all(u1(t, s[t], r[s[t]]) >= max(u1(t, m, r[m]) for m in msgs)
             ↪ for t in types):
                out.append((s, r, beliefs))
    return out


# Gibbons Fig. 4.2.2 (p. 189): (sender, receiver) payoffs; t1, t2 equally likely
G422 = {("t1", "L", "u"): (1, 3), ("t1", "L", "d"): (4, 0), ("t1", "R", "u"): (2, 1),
 ↪ ("t1", "R", "d"): (0, 0),
        ("t2", "L", "u"): (2, 4), ("t2", "L", "d"): (0, 1), ("t2", "R", "u"): (1, 0),
 ↪ ("t2", "R", "d"): (1, 2)}
g = pure_pbe(("t1", "t2"), half, ("L", "R"), ("u", "d"),
             lambda t, m, a: G422[(t, m, a)][0], lambda t, m, a: G422[(t, m, a)][1])
found = {(tuple(s.values()), tuple(r.values())): b for s, r, b in g}
assert set(found) == {(("L", "L"), ("u", "d")), (("R", "L"), ("u", "u"))}       # pp.
 ↪ 188-190
assert found[(("L", "L"), ("u", "d"))]["R"] == (0, Fr(2, 3))                    # q
 ↪ <= 2/3
```

---

4 · The MBA game: matrix, BNE and PBE → L08-05-mba-game

The 16-cell Bayesian normal form at $p = 1/4$ (p. 322), its two BNE, and the solver's verdict: (DU, BM) with $\mu_U = 0,\ \mu_D = 1$; (UU, BB) with $\mu_U = 1/4$ and $\mu_D \in [0, 3/8]$.

```
# Tadelis Sec. 16.1 (pp. 319-323): wages and study costs, employer profits
def mba(wM=10, wB=6, cH=2, cL=5):
    """Sender/receiver payoffs of the MBA game; returns (u1, u2)."""
    wage, cost = {"M": wM, "B": wB}, {"H": cH, "L": cL}
    prof = {("H", "M"): 10, ("H", "B"): 5, ("L", "M"): 0, ("L", "B"): 3}
    u1 = lambda t, m, a: wage[a] - (cost[t] if m == "D" else 0)
    u2 = lambda t, m, a: prof[(t, a)]
    return u1, u2


u1, u2 = mba()
quarter = Fr(1, 4)
# the 4x4 Bayesian normal form at p = 1/4 (p. 322)
def nf(s1, s2, p=quarter):
```

```
    """s1 = (H's message, L's message); s2 = (response to U, response to D)."""
    resp = {"U": s2[0], "D": s2[1]}
    return (p * u1("H", s1[0], resp[s1[0]]) + (1 - p) * u1("L", s1[1], resp[s1[1]]),
            p * u2("H", s1[0], resp[s1[0]]) + (1 - p) * u2("L", s1[1], resp[s1[1]]))


assert nf("UD", "MB") == (Fr(13, 4), Fr(19, 4))                                    #
 ↪  (3.25, 4.75)
ROWS, COLS = ["UU", "UD", "DU", "DD"], ["MM", "MB", "BM", "BB"]
TABLE = {"UU": [(10, 2.5), (10, 2.5), (6, 3.5), (6, 3.5)], "UD": [(6.25, 2.5), (3.25,
 ↪  4.75), (5.25, 1.25), (2.25, 3.5)],
         "DU": [(9.5, 2.5), (8.5, 1.25), (6.5, 4.75), (5.5, 3.5)], "DD": [(5.75,
          ↪  2.5), (1.75, 3.5), (5.75, 2.5), (1.75, 3.5)]}
assert all(tuple(map(float, nf(r, c))) == TABLE[r][j] for r in ROWS for j, c in
 ↪  enumerate(COLS))
# (DU, BB): 1/4*4 + 3/4*6 = 5.5; the book prints 4.5 (p. 322), which changes no best
 ↪  response
assert sorted(pure_nash(ROWS, COLS, nf)) == [("DU", "BM"), ("UU", "BB")]

pbe = pure_pbe(("H", "L"), quarter, ("U", "D"), ("M", "B"), u1, u2)
eqs = {(s["H"] + s["L"], r["U"] + r["D"]): b for s, r, b in pbe}
assert set(eqs) == {("DU", "BM"), ("UU", "BB")}
assert eqs[("DU", "BM")] == {"U": 0, "D": 1}                                        #
 ↪  separating, p. 322
assert eqs[("UU", "BB")] == {"U": quarter, "D": (0, Fr(3, 8))}                      #
 ↪  pooling, mu_D <= 3/8
```

---

5 · The intuitive criterion → L08-08-intuitive-criterion

Tadelis Def. 16.1 for two types. Checked on the MBA game (pooling fails, separating survives), on Gibbons's Fig. 4.2.2 pooling equilibrium, and on Gibbons's Beer–Quiche pooling on Quiche (p. 239).

```
def intuitive_fails(eq, types, prior, msgs, acts, u1, u2):
    """Tadelis Def. 16.1 for two types: True if some off-path message breaks the PBE
     ↪  `eq`.

    A type is excluded from message m when even the best reply to m (over all
 ↪  beliefs) leaves it
    strictly below its equilibrium payoff. If the remaining types are a nonempty
 ↪  proper subset and
    one of them gains for sure once the receiver believes only they could send m, the
 ↪  PBE fails.
    """
    s, r, _ = eq
    star = {t: u1(t, s[t], r[s[t]]) for t in types}
    t1, t2 = types
    for m in msgs:
        if m in s.values():
            continue
        any_br = [a for a in acts if br_interval(u2, m, a, acts, t1, t2) is not None]
```

```
        keep = [t for t in types if max(u1(t, m, a) for a in any_br) >= star[t]]
        if not keep or len(keep) == len(types):
            continue
        mu = Fr(1) if keep == [t1] else Fr(0)                    # belief on the
 ↪  kept type only
        br = [a for a in acts if (iv := br_interval(u2, m, a, acts, t1, t2)) and
 ↪  iv[0] <= mu <= iv[1]]
        if any(min(u1(t, m, a) for a in br) > star[t] for t in keep):
            return True
    return False


by_key = {(s["H"] + s["L"], r["U"] + r["D"]): (s, r, b) for s, r, b in pbe}
assert intuitive_fails(by_key[("UU", "BB")], ("H", "L"), quarter, ("U", "D"), ("M",
 ↪  "B"), u1, u2)       # p. 334
assert not intuitive_fails(by_key[("DU", "BM")], ("H", "L"), quarter, ("U", "D"),
 ↪  ("M", "B"), u1, u2)
# Gibbons Fig. 4.2.2: R is equilibrium-dominated for t2 only (best 1 < 2); believed
 ↪  to be t1,
# the Receiver plays u after R and t1 gets 2 > 1, so the pooling PBE on L fails the
 ↪  criterion
gk = {(tuple(s.values()), tuple(r.values())): (s, r, b) for s, r, b in g}
G1 = lambda t, m, a: G422[(t, m, a)][0]; G2 = lambda t, m, a: G422[(t, m, a)][1]
assert intuitive_fails(gk[(("L", "L"), ("u", "d"))], ("t1", "t2"), half, ("L", "R"),
 ↪  ("u", "d"), G1, G2)

# Beer and Quiche, Gibbons Fig. 4.4.3 (p. 238): wimpy w.p. 0.1; payoffs (Sender,
 ↪  Receiver)
BQ = {("w", "Q", "N"): (3, 0), ("w", "Q", "D"): (1, 1), ("w", "B", "N"): (2, 0),
 ↪  ("w", "B", "D"): (0, 1),
      ("s", "Q", "N"): (2, 0), ("s", "Q", "D"): (0, -1), ("s", "B", "N"): (3, 0),
 ↪  ("s", "B", "D"): (1, -1)}
B1 = lambda t, m, a: BQ[(t, m, a)][0]; B2 = lambda t, m, a: BQ[(t, m, a)][1]
bq = pure_pbe(("w", "s"), Fr(1, 10), ("Q", "B"), ("N", "D"), B1, B2)
bk = {(s["w"] + s["s"], r["Q"] + r["B"]): (s, r, b) for s, r, b in bq}
assert bk[("QQ", "ND")][2]["B"] == (half, 1)                        # pooling on
 ↪  Quiche needs q >= 1/2, p. 238
assert intuitive_fails(bk[("QQ", "ND")], ("w", "s"), Fr(1, 10), ("Q", "B"), ("N",
 ↪  "D"), B1, B2)     # p. 239
# (the full equilibrium set of this game is Tadelis Exercise 16.8, p. 338: left as
 ↪  practice)
```

---

6 · Spence with linear costs, and the hybrid equilibria → L08-06-spence-job-market

M&R's numbers: separating $[1.5, 3]$, pooled wage 29, pooling up to 0.45, H nets 35, $p^* = 0.5$; Gibbons (4.2.8)–(4.2.10) for the hybrid at $e_h = 1$.

```
def spence(tH, tL, p, cH, cL):
    """Two-type Spence model, unproductive education, cost c_t * e (M&R pp. 155-156).
```

```
    Returns the separating interval for e_H (L's education is 0), the pooled wage,
 ↪  the largest
    pooling education sustained by the belief 'any other e means L', H's
 ↪  least-cost-separating
    payoff, and the prior above which H prefers a world without signaling.
    """
    assert tH > tL and 0 < p < 1 and cH < cL, "single crossing needs c_H < c_L"
    lo, hi = (tH - tL) / cL, (tH - tL) / cH          # IC_L: tL >= tH - cL e ; IC_H:
 ↪  tH - cH e >= tL
    wbar = p * tH + (1 - p) * tL
    return dict(sep=(lo, hi), wbar=wbar, ep_max=(wbar - tL) / cL,
                H_lcs=tH - cH * lo, p_star=1 - cH / cL)


mr = spence(50, 20, 0.30, 10, 20)
assert mr["sep"] == (1.5, 3.0)                  # M&R: "E_H a little bit more than
 ↪  1.5"; E_H = 2 works
assert abs(mr["wbar"] - 29) < 1e-12             # (.3 x 50) + (.7 x 20) = 29
assert abs(mr["ep_max"] - 0.45) < 1e-12 and mr["H_lcs"] == 35 and mr["p_star"] == 0.5
assert 50 - 20 * 2 == 10 < 20 and 50 - 10 * 2 == 30 > 20      # M&R's E_H = 2 check,
 ↪  p. 156


def hybrid(tH, tL, p, cH, cL, eh):
    """Gibbons (4.2.8)-(4.2.10): L mixes between 0 and eh; returns (wage, belief r,
     ↪  L's prob pi)."""
    wh = tL + cL * eh                            # L indifferent: tL - 0 = wh - cL eh
 ↪  (4.2.9)
    r = (wh - tL) / (tH - tL)                    # wh = r tH + (1 - r) tL
 ↪  (4.2.10)
    pi = p * (1 - r) / (r * (1 - p))             # invert r = p / (p + (1 - p) pi)
 ↪  (4.2.8)
    return wh, r, pi


wh, r, pi = hybrid(50, 20, 0.30, 10, 20, 1.0)
assert (wh, round(r, 4), round(pi, 4)) == (40, 0.6667, 0.2143)
assert hybrid(50, 20, 0.30, 10, 20, 1.5)[2] == 0                    # eh ->
 ↪  least-cost: pure separation
assert abs(hybrid(50, 20, 0.30, 10, 20, 0.45)[2] - 1) < 1e-12         # eh -> pooling
 ↪  bound: L always pools
```

---

7 · Limit pricing: monopoly, Cournot, separating and pooling bounds → L08-07-limit-pricing

Every number of §16.2: Steps 1–2 as linear systems, the separating interval $[2.62, 3.32]$, the Bayesian Cournot quantities $(19/12, 13/12, 5/6)$, the pooling interval $[1.493, 1.917]$, and the intuitive-criterion inequality (16.3).

```python
import numpy as np

# Tadelis Sec. 16.2 (pp. 323-332): P = 5 - Q, c2 = 2, entry cost 1/2, c1 in {1 (L), 2 (H)}
mono = lambda c: ((5 - c) / 2, ((5 - c) / 2) ** 2)                # Step 1: q and profit
assert mono(1) == (2, 4) and mono(2) == (1.5, 2.25)
# Step 2 (complete information Cournot), solved as a linear system of the two FOCs
def cournot(c1, c2=2):
    q = np.linalg.solve([[2, 1], [1, 2]], [5 - c1, 5 - c2])
    price = 5 - q.sum()
    return q, (price - c1) * q[0], (price - c2) * q[1] - 0.5
qH, v1H, v2H = cournot(2); qL, v1L, v2L = cournot(1)
assert np.allclose(qH, [1, 1]) and abs(v1H - 1) < 1e-12 and abs(v2H - 0.5) < 1e-12
assert np.allclose(qL, [5 / 3, 2 / 3]) and abs(v1L - 25 / 9) < 1e-12 and abs(v2L + 1 / 18) < 1e-12

# best deviation payoffs when any deviation triggers entry with q2 = 1
dev_H = 2.25 + 1.0          # monopoly now, Cournot reply (2 - q) q at q = 1 next period
dev_L = 4.0 + 2.25          # monopoly now, reply (3 - q) q at q = 1.5 next period
# separating: H must not mimic, L must not deviate (pp. 328-329)
sep_lo = (3 + 5 ** 0.5) / 2                                       # (3 - q) q + 2.25 <= 3.25
sep_hi = 2 + 1.75 ** 0.5                                          # (4 - q) q + 4 >= 6.25
assert abs((3 - sep_lo) * sep_lo + 2.25 - dev_H) < 1e-12 and abs((4 - sep_hi) * sep_hi + 4 - dev_L) < 1e-12
assert (round(sep_lo, 2), round(sep_hi, 2)) == (2.62, 3.32)
assert (5 - 2 - 2) * 2 + 2.25 == 4.25 > 3.25                      # q1L = 2 fails: H mimics (p. 328)

# pooling: Bayesian Cournot after entry with belief 1/2 (pp. 330-331)
A = np.array([[2, 0, 1], [0, 2, 1], [0.5, 0.5, 2]])               # FOCs in (qL, qH, q2)
qL2, qH2, q2 = np.linalg.solve(A, [4, 3, 3])
assert np.allclose([qL2, qH2, q2], [19 / 12, 13 / 12, 5 / 6])
v2_pool = q2 * (3 - q2 - (qL2 + qH2) / 2) - 0.5
assert abs(v2_pool - 7 / 36) < 1e-12                              # > 0: entry. The book prints 7/18 (p. 331)
roots = lambda b, c: sorted(np.roots([1, -b, c]).real)            # q^2 - b q + c = 0
pH = roots(3, 3.25 - qH2 ** 2); pL = roots(4, 6.25 - qL2 ** 2)
assert np.allclose(pH, [13 / 12, 23 / 12]) and np.allclose(pL, [1.4931, 2.5069], atol=1e-4)
pool = (max(pH[0], pL[0]), min(pH[1], pL[1]))
assert (round(pool[0], 3), round(pool[1], 3)) == (1.493, 1.917)   # p. 332
# intuitive criterion at q = 3 (p. 334): H loses for sure, L gains if believed
assert (5 - 3 - 2) * 3 + 2.25 < 3.25 and (5 - 3 - 1) * 3 + 4 == 7 > 4 + qL2 ** 2
```

---

8 · Sequential equilibrium: trembles pin the belief → L08-03-sequential-equilibrium

Fig. 15.4: whatever the tremble, Bayes gives $\mu_2(x_1) = 1/2$, so R (which needs $\mu > 2/3$) is never sequentially rational.

```
# Tadelis Fig. 15.4 (p. 312): consistency pins mu2(x1) = 1/2 for every tremble
for eps in [Fr(1, 10), Fr(1, 1000), Fr(1, 10 ** 6)]:
    post = posterior({"x1": half, "x2": half}, {"x1": eps, "x2": eps})    # 1 cannot
 ↪  condition on Nature
    assert post["x1"] == half
mu = half
assert 3 * mu < 2                                    # R pays 3 mu, L pays 2: R is not
 ↪  sequentially rational
assert all(3 * m > 2 for m in [Fr(7, 10), Fr(9, 10)])   # the PBE (U, R) needs
 ↪  mu2(x1) > 2/3
print("L08 reference implementation: all checks pass.")
```

Links

- Figures: `figures/make_figures.py` (run `python Notas/L08/figures/make_figures.py`), whose asserts re-check the plotted numbers.
- Companions and their checks: L08-00-map §Interactive companions.
- Rules: rules/game-theory item 12 ("verify by checking no player has a profitable deviation" — which is what `pure_pbe` does exhaustively).